

LCM-UNet: A Reparameterizable Local-Compensated Mamba U-Net for Skin Lesion Segmentation

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Abstract

State space models have made it possible to build skin lesion segmentation networks with only a few tens of thousands of parameters, which is attractive for point-of-care dermatological screening on low-cost hardware. In such ultra-compact designs the local pathway is usually reduced to a single depthwise-separable branch operating on a quarter of the channels, so the network retains very little capacity for the fine spatial detail that determines boundary quality. Enlarging that branch is not an acceptable remedy, because any additional operator is paid for at inference time in a regime where the entire model costs less than 0.06 GFLOPs. This article presents LCM-UNet, which strengthens the local pathway without changing the deployed computational graph. The proposed Residual Dilated Local Adapter (RDLA) attaches a zero-initialized, dilation-two depthwise 3×3 convolution in parallel with the existing depthwise kernel of each global-local block, and sums its response into the pre-activation depthwise output. Because both branches are bias free and are combined before the first normalization, the pair is merged exactly into a single sparse kernel after training, following the dilated reparameterization equivalence established for large-kernel convolutional backbones. Our contribution is not that equivalence but its behaviour at a scale where it has not been characterized: host kernels of 25 or 49 weights per channel, inside a 34k-parameter state space segmentation network. A minor refinement replaces the single shared outer skip coefficient with five stage-specific scalars. Across three seeds on ISIC 2017 and ISIC 2018, LCM-UNet improves over a locally retrained reference backbone by 0.48 and 0.37 percentage points in IoU and 0.29 and 0.22 points in DSC, with lower HD95 and ASSD on both datasets. On PH², an external dataset seen by neither model during training, evaluating the reference backbone under the same protocol turns the transfer measurement into a controlled comparison: across six checkpoints per model, LCM-UNet reaches $86.37 \pm 0.29\%$ IoU against $85.91 \pm 0.29\%$ for the reference, a gain that closely tracks the in-distribution result. The improvements are small in absolute terms but statistically well supported: the seed-paired run-level test is significant on both datasets, the unpaired three-seed test additionally reaches significance on ISIC 2017, and the paired image-level Wilcoxon analysis is significant at $p < 10^{-4}$ on 645 and 808 images with rank-biserial correlations of 0.19–0.20. We report these figures as evidence of a consistent and statistically robust shift rather than a substantial advance in accuracy. Fused and unfused models produce identical thresholded masks to numerical tolerance, and the reparameterized network uses 34,180 parameters and 0.05974 GFLOPs, four parameters more than the reference at identical computational cost. The result calibrates how much reparameterization-based local compensation is worth in the ultralight state space regime.

Index Terms

Skin lesion segmentation, state space model, Mamba, lightweight neural network, structural reparameterization, dilated convolution, U-Net.

I. INTRODUCTION

Melanoma remains one of the most lethal forms of skin cancer, and the probability of successful treatment depends strongly on how early a suspicious lesion is identified [1], [2]. The five-year relative survival rate for melanoma detected while still localized to the skin exceeds 99%, against roughly 76% once it has spread regionally and roughly 35% once it has metastasized to distant sites [3]; that gap is the clinical case for automated tools that support earlier and more consistent screening rather than a claim that segmentation alone changes outcomes. Dermoscopy is the routine imaging modality for this task, and automatic delineation of the lesion boundary is an essential preprocessing step for downstream measurements such as asymmetry, border irregularity and diameter. Manual delineation is slow, operator dependent, and difficult to reproduce across clinics, which has motivated a long line of work on automated segmentation of dermoscopic images.

Encoder-decoder convolutional networks, and in particular the U-Net family, established the dominant design pattern for this problem [4]–[6]. Later work replaced or augmented the convolutional backbone with self-attention in order to model long-range spatial dependence [7], [8]. Attention-based models improved global context aggregation but introduced a quadratic dependence on sequence length, which is a serious obstacle when the target platform is a mobile device or an embedded system

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in a primary-care setting. A parallel line of research therefore focused on architectural compression, producing segmentation networks with parameter counts in the range of 0.05–0.3 million [9]–[11].

The recent adoption of structured state space models, and specifically the selective scan mechanism of Mamba [12], changed the trade-off again. Because a selective scan has linear complexity in sequence length, Mamba-based encoders can aggregate global context at a cost that convolutional and attention backbones cannot match at equal parameter budget [13], [14]. Applied to medical image segmentation, this produced a family of very compact networks [15]–[18], culminating in designs that split the channel dimension into parallel global and local groups so that a state space branch and a lightweight convolutional branch operate side by side on disjoint channel subsets [18], [19]. UltraLBM-UNet [19] is representative of the current operating point in this regime: it reaches competitive accuracy on ISIC 2017, ISIC 2018 and PH² with roughly 0.034 million parameters and 0.060 GFLOPs.

The channel-splitting strategy that makes these networks small also creates a specific weakness. When a block of width C is divided into four groups, the local pathway receives only $C/4$ channels and is implemented as a single depthwise-separable convolution. In the deepest stages of an ultra-light network this amounts to eight to sixteen depthwise channels. Lesion boundaries in dermoscopic images are frequently low in contrast, irregular, and partially occluded by hair or air bubbles [20], [21], and these are precisely the cases in which a narrow local pathway with a single fixed sampling grid struggles. The obvious remedies are unattractive. Widening the local group takes channels away from the state space branch; adding a second convolutional branch, an attention module or a multi-scale aggregation block permanently increases inference cost, which contradicts the design objective of the whole model class.

This observation motivates a different question: *can the local pathway be given extra representational capacity during optimization only, and then be collapsed back into the original operator before deployment?* Structural reparameterization provides the mechanism. Training-time multi-branch topologies that are linear and bias-free before their first normalization layer can be folded exactly into a single equivalent kernel, a principle established for plain convolutional backbones [22]–[24] and later extended to large-kernel networks by re-expressing dilated small-kernel branches as sparse large kernels [25], [26]. This machinery has been developed and evaluated almost exclusively on dense classification and detection backbones with millions of parameters. Whether it remains useful, or even measurable, in a segmentation network two orders of magnitude smaller, where the host depthwise kernel has fewer than two hundred weights, has not been characterized.

We address that question with LCM-UNet, a local-compensated Mamba U-Net built on the UltraLBM-UNet layout.

The contribution is the *Residual Dilated Local Adapter* (RDLA). In each global–local block that uses a large local kernel, a zero-initialized depthwise 3×3 convolution with dilation two is placed in parallel with the existing depthwise kernel, and its output is added to the pre-activation depthwise response. Placement before the first batch normalization is what makes exact fusion possible; zero initialization ensures that the adapter starts as an exact identity with respect to the parent network, so training begins from the reference function and the branch is grown only if the data supports it. After training, the 3×3 dilated kernel is embedded into the sparse grid of the host 5×5 or 7×7 kernel and summed with it, after which the adapter is deleted.

We also apply one minor refinement, *stage-specific outer skip scaling*. The parent network applies a single learnable scalar to all five encoder–decoder skip connections, which forces the shallowest and deepest levels to share one mixing ratio despite differing by an order of magnitude in resolution and a factor of eight in width. Removing that parameter sharing costs four scalars and introduces no new operator type. We treat it as a design refinement rather than as a second contribution, because its isolated effect is not distinguishable from zero in our ablation (Section V-E), and we report it in that spirit.

The contributions of this article are as follows.

- 1) A lightweight adapter that costs nothing at deployment. RDLA is a zero-initialized dilated depthwise branch added to the local pathway of global–local state space blocks during training, and folded back into the host kernel by an exact closed-form identity afterwards. The training-form network carries 432 extra weights; the model that ships carries none. The underlying fusion identity builds on prior work [26], and Section III-E spells out what is different here, mechanism by mechanism, from the closest existing constructions.
- 2) A test of that idea at a scale where it had not been tried. Prior reparameterization studies operate on host kernels with hundreds to thousands of weights; here the host kernels hold 25 or 49 weights per channel, split across groups of eight to sixteen channels inside a 34k-parameter block that shares its output with a selective-scan branch. We report what the technique is worth in that regime rather than assuming results from much larger backbones carry over.
- 3) An evaluation designed to be checked, not just read. Three-seed results on ISIC 2017 and ISIC 2018, external validation on PH², boundary-sensitive metrics (Boundary Dice, HD95, ASSD), run-level and paired image-level significance tests with effect sizes and confidence intervals, and a numerical fusion-equivalence audit for every trained checkpoint. We report effect sizes as small where they are small, including cases that do not reach significance.

The remainder of this article is organized as follows. Section II reviews lightweight segmentation, state space vision models and structural reparameterization. Section III describes the architecture, the adapter, its reparameterization, and the stage-specific skip scaling. Section IV details datasets, protocol and metrics. Section V presents the main comparison, ablation, boundary analysis, statistical testing and efficiency audit. Section VI discusses what the results do and do not establish, and Section VII concludes.

II. RELATED WORK

A. Classical and Model-Based Segmentation

Before learned features: thresholding, active contours, and hybrid pipelines. Automated dermoscopic border detection predates deep learning by two decades, and it is worth recalling what that literature established, because several of its findings still shape how the field frames the problem today. Silveira *et al.* [27] compared six representative families head to head on the same image set: gradient vector flow snakes, level-set evolution, adaptive thresholding, an adaptive snake, an expectation-maximization level-set variant, and a fuzzy-logic split-and-merge scheme. The adaptive snake and the semi-supervised expectation-maximization level-set method gave the closest agreement with dermatologist-drawn borders, while the fuzzy split-and-merge algorithm was the strongest fully automatic method, and the comparison made explicit a trade-off that recurs throughout this article: methods that use additional supervision or manual initialization outperform fully automatic ones, and the practical target has always been to close that gap without adding supervision. Celebi *et al.* [28] approached the same problem through ensembles of thresholding rules combined by simple voting, which improved robustness to the uneven illumination and color variation that single-threshold methods struggled with; the observation that no single fixed rule generalizes across the full range of skin tones, hair density and acquisition hardware anticipated exactly the appearance shift our own dataset figure documents (Fig. 5) and that motivates learned, data-driven boundary models. Garnavi *et al.* [29] pushed the same hybrid-thresholding idea further by first searching for the color channel combination that best separated lesion from skin on a per-image basis before thresholding, which is, in effect, a hand-engineered precursor to the per-image adaptivity that a trained convolutional or state-space encoder now provides implicitly. Two conclusions from this line of work carry over directly to the present study. First, boundary quality, not region-level discrimination, was already understood to be the hard part of dermoscopic segmentation before any of it was learned, which is why Section V-G reports Boundary Dice, HD95 and ASSD as first-class results rather than as an afterthought to IoU and DSC. Second, hybrid pipelines that combined several weak, inexpensive rules routinely outperformed any single rule alone, a pattern echoed here at the architectural level: the global-local block underlying LCM-UNet already combines two mechanisms of different character (a selective scan and a depthwise convolution), and the adapter studied in this article adds a third, narrow, largely automatic one to the weakest of the two.

B. U-Net Variants for Skin Lesion Segmentation

Encoder-decoder baselines and their attention-based successors. U-Net [4] introduced the symmetric encoder-decoder with skip connections that still underpins most medical segmentation systems. UNet++ [5] densified the skip pathway with nested convolutional bridges, and Attention U-Net [6] gated skip features so that irrelevant background responses are suppressed before fusion. A large body of applied work has adapted this template specifically to dermoscopy. MultiResUNet [30] replaced the plain skip with residual multi-scale paths; separable-UNet [31] substituted depthwise-separable convolutions throughout the encoder; adversarially trained variants such as the dense-residual network of Tu *et al.* [32] and the attention-based DenseUNet of Wei *et al.* [33] added a discriminator to sharpen predicted boundaries; and ensemble [34] and improved-CNN [35] formulations traded computation for accuracy. More recent designs combine attention with multi-scale aggregation, as in the visual-attention-network cross-attention scheme of Liu *et al.* [21] and the progressive class-wise attention of Naveed *et al.* [36]. For dermoscopic images specifically, the difficulty is less about semantic ambiguity than about boundary quality: lesions are large, roughly convex, and usually well centred, so the region-level decision is comparatively easy while the contour is frequently low in contrast and disturbed by hair, rulers, and illumination gradients. This is the reason so much of the applied literature targets the contour rather than the region, and it is the reason our own evaluation reports boundary-sensitive measures alongside overlap. Transformer hybrids such as TransUNet [7] and Swin-UNet [8] improved global context modelling and reported strong results, but their computational profile is dominated by attention, which scales quadratically with the token count and is therefore poorly suited to on-device screening.

C. Lightweight and Efficient Segmentation Networks

Reducing parameters and FLOPs without abandoning the U-shape. A separate research direction accepted the U-shaped topology and attacked its cost. DSNet [20] was an early demonstration that depthwise-separable projection alone removes most of the parameter cost of a dermoscopic segmenter without collapsing accuracy. SLSNet [37] pursued the same objective through one-dimensional kernel factorization inside a generative adversarial framework, motivated explicitly by the memory limits of dermoscopy hardware. UNeXt [11] replaced deep convolutional stages with tokenized MLP blocks. MALUNet [9] combined several lightweight attention modules to reach roughly 0.18 M parameters, and EGE-UNet [10] pushed further to about 0.05 M by combining group-wise multi-axis attention with a grouped aggregation bridge. ELA-Net [38] reached 0.459 M with atrous convolution and attention, and UCM-Net [39] continued the trend with mixed convolutional and multilayer-perceptron stages. These networks demonstrated that dermoscopic segmentation tolerates aggressive compression, but they also converged on a common structure in which the local convolutional pathway is extremely narrow, which is the pressure point that the present work targets.

D. State Space Models in Vision and Medical Imaging

Linear-complexity global context as an alternative to attention. Structured state space sequence models [40] and their selective variant, Mamba [12], provide sequence mixing with linear complexity. Vision Mamba [13] and VMamba [14] adapted the mechanism to images through bidirectional and multi-directional scans. In medical segmentation, VM-UNet [15], U-Mamba [41], LightM-UNet [16], Swin-UMamba [17] and MedMamba [42] placed state space blocks in encoder, decoder, or both. UltraLight VM-UNet [18] made the decisive efficiency step by splitting the channel dimension and processing several channel groups with parallel Mamba branches, which decouples parameter count from feature width. UltraLBM-UNet [19], a preprint at the time of writing, extends this idea into a global-local multi-branch block in which two channel groups are processed by a weight-shared bidirectional Mamba, one group by a depthwise-separable convolution, and one group by an identity path, reaching roughly 0.034 M parameters and 0.060 GFLOPs. We adopt UltraLBM-UNet as the structural reference of this work, and we retain its Mamba configuration, weight sharing, and internal residual scaling unchanged so that the effect of our modifications is isolated. The field has continued to move quickly since: MUCM-Net [43] added a Mamba branch to the same authors' earlier UCM-Net [39], and LiteMamba-Bound [44] paired a channel-attention dual-Mamba block with a boundary-aware active-contour loss aimed directly at the contour-quality weakness this article also targets, at roughly 0.96 M parameters – almost thirty times the budget considered here. The loss-function route and the architectural route explored in this article are complementary rather than competing: a boundary-aware loss changes what the optimizer is rewarded for, while RDLA changes what the local branch is capable of representing in the first place, and nothing in the present design precludes combining the two.

E. Structural Reparameterization

Training-time branches that vanish at inference. Structural reparameterization exploits the linearity of convolution and batch normalization before the first nonlinearity: a multi-branch training topology can be folded into one equivalent kernel. ACNet [24] used asymmetric 1×3 and 3×1 branches, RepVGG [22] folded identity and 1×1 branches into plain 3×3 convolutions, and the Diverse Branch Block [23] generalized the construction to a richer branch set. MobileOne [45] transferred the idea to mobile backbones. For large kernels, RepLKNet [25] added a parallel small kernel to stabilize optimization, and UniRepLKNet [26] introduced the Dilated Reparam Block, which observes that a dilated small-kernel convolution is equivalent to a non-dilated convolution with a larger *sparse* kernel, and can therefore be merged into the host large kernel for inference. Our RDLA follows this equivalence directly: it is a single-branch, depthwise, zero-initialized instance of the dilated reparameterization principle. Three points distinguish the present study from the prior art. First, all of the above operate on dense classification or detection backbones with millions of parameters, where a host kernel has thousands of weights; our host kernels have 25 or 49 weights per channel inside a 34k-parameter segmentation network, a regime where the technique has not been evaluated and where its benefit is not guaranteed. Second, the adapter is inserted into the convolutional group of a *global-local state space block* rather than into a plain convolutional stage, so it interacts with a Mamba branch that shares the same block output. Third, we initialize the adapter to zero, in the spirit of residual-scaling schemes [46], [47] and zero-initialized adaptation layers [48], [49], so that the training-form network is function-identical to the reference model at step zero and departs from it only under gradient pressure.

F. Dilated Convolution and Skip-Connection Design

Enlarging the sampling grid and controlling encoder-decoder fusion. Dilated convolution [50], [51] enlarges the receptive field without additional weights and is a standard component of segmentation decoders, though it is normally deployed as a persistent inference operator, often in multi-rate pyramids. Our use is different in kind: the dilated branch exists only during training and is removed afterwards, so it contributes sampling diversity to optimization rather than an inference-time multi-scale module. On the fusion side, skip connections have been refined by dense nesting [5], attention gating [6], and learned weighting. Ultra-light Mamba U-Nets instead apply a single learnable scalar to all skip levels [19], which is the most economical possible choice. We relax only the parameter sharing of that scalar and deliberately avoid convolutional adapters or gating networks, so that the four additional parameters cannot be confused with added capacity.

G. Positioning

Table I summarizes the closest works, and Table III in Section III-E gives a mechanism-level comparison against the reparameterization constructions specifically. The gap addressed here is narrow: ultra-light global-local state space segmenters have a local branch that is too narrow to refine boundaries, and the reparameterization tools that could enlarge it for free at inference have not been characterized at this scale. LCM-UNet combines the two, and the experiments in Section V quantify the resulting benefit including its limits.

Stepping back across the five subsections above, the field has moved through three largely sequential phases on this specific problem. Classical pipelines (Section II-A) established that boundary quality, not region-level discrimination, is the hard part of dermoscopic segmentation, but had no mechanism to learn from data what a fixed thresholding or active-contour rule could

TABLE I: Summary of Closely Related Work

Method	Year	Family	Params (M)	Core idea and relation to this work
U-Net [4]	2015	CNN	31.0	Symmetric encoder–decoder with skip fusion; the topology adopted here.
UNet++ [5]	2018	CNN	9.16	Nested dense skip pathway; adds inference cost at every level.
Attention U-Net [6]	2018	CNN	34.9	Gated skips; a learned gate rather than a scalar coefficient.
TransUNet [7]	2021	Hybrid	105	Transformer encoder for global context; quadratic attention cost.
Swin-UNet [8]	2022	Transformer	27.2	Shifted-window attention; still far above the mobile budget.
UNeXt [11]	2022	MLP	1.47	Tokenized MLP stages replace deep convolutions.
MALUNet [9]	2022	CNN	0.178	Multiple lightweight attention modules; common evaluation protocol.
EGE-UNet [10]	2023	CNN	0.053	Group multi-axis attention and grouped aggregation bridge.
DSNet [20]	2020	CNN	5.83	Depthwise-separable projection for dermoscopic segmentation; early compact baseline.
ELA-Net [38]	2024	CNN	0.459	Atrous convolution with attention at sub-million scale.
UCM-Net [39]	2024	CNN+MLP	0.049	Mixed convolution–MLP stages for compact dermoscopic segmentation.
VM-UNet [15]	2024	SSM	27.4	First full Mamba U-Net for medical segmentation.
LightM-UNet [16]	2024	SSM	1.09	Lightweight Mamba assistance inside a compact U-Net.
UltraLight VM-UNet [18]	2024	SSM	0.045	Channel-split parallel Mamba; decouples cost from feature width.
UltraLBM-UNet [19]	2025	SSM	0.034	Global–local block with weight-shared bidirectional Mamba; structural reference of this work.
RepVGG [22]	2021	Reparam.	–	Multi-branch training topology folded into plain convolutions.
UniRepLKNet [26]	2024	Reparam.	–	Dilated small kernels merged into a large kernel; the equivalence RDLA builds on, previously shown only on large backbones.
LCM-UNet (this work)	–	SSM+Reparam.	0.034	Zero-initialized dilated depthwise adapter fused into the local branch, plus five stage-specific skip scalars; no inference-time operator is added.

Parameter counts for prior methods are the values reported by their authors and are listed for orientation only; they were not re-measured here. “–” denotes a backbone technique rather than a segmentation model.

not express. Learned encoder–decoders, from U-Net through its attention and transformer successors, replaced the fixed rule with a learned one and, in the lightweight branch of that literature, drove parameter counts down by roughly three orders of magnitude relative to the transformer hybrids, at the structural cost of shrinking the local convolutional pathway to a handful of channels. State space models restored global context at that reduced budget without reintroducing quadratic cost, which is why every architecture compared in Table XIV is Mamba-based; what none of them revisited is whether the now-narrow local pathway itself still has enough capacity, since the channel-splitting design that made the global branch cheap also made the local branch the narrowest component in the network. Structural reparameterization is, in this framing, not a fourth phase but a tool borrowed from a different line of work entirely (large-kernel convolutional backbones) and applied to the specific pressure point the second and third phases created and left unaddressed. The contribution of this article is accordingly best read as connective rather than as a new architectural family: it identifies which existing tool fits the gap the literature’s own trajectory produced, and reports, honestly, how much that fit is worth at the scale the trajectory arrived at.

III. METHODOLOGY

A. Overview and Design Constraints

LCM-UNet is a five-level U-shaped encoder–decoder that follows the stage layout of UltraLBM-UNet [19]. The design is governed by one constraint that we impose on every component: *the operator set of the deployed model must not change*. Any module that is added for training must either be removable by an exact algebraic identity or cost so little that it does not alter the deployment profile. This constraint rules out extra attention blocks, auxiliary decoders, multi-rate pyramids, and convolutional skip adapters, and it is what makes the two proposed components as small as they are.

Fig. 1 shows the complete network. Given an input image $X \in \mathbb{R}^{3 \times H \times W}$ with $H = W = 256$, six encoder stages produce features of width $\{8, 16, 24, 32, 48, 64\}$. Stages E1–E3 are plain convolutional stages. Stage E4 is a local multi-branch perception (LMBP) block with a 3×3 local kernel. Stages E5 and E6 are global–local multi-branch perception (GLMBP) blocks with local kernels of size 5 and 7. Each of E1–E5 is followed by 2×2 max pooling, and E6 operates at the bottleneck resolution of 8×8 . The decoder mirrors this arrangement with GLMBP blocks at D1 and D2, an LMBP block at D3, convolutional stages at D4 and D5, and a 1×1 convolution producing a single-channel logit map. Upsampling is bilinear with `align_corners=True`. Table II lists the stages, channel transitions and local kernel sizes.

TABLE II: Stage Configuration of LCM-UNet

Stage	Operator	Channels	Local K	RDLA
E1	Conv	$3 \rightarrow 8$	3	–
E2	Conv	$8 \rightarrow 16$	3	–
E3	Conv	$16 \rightarrow 24$	3	–
E4	LMBP	$24 \rightarrow 32$	3	–
E5	GLMBP	$32 \rightarrow 48$	5	✓
E6	GLMBP	$48 \rightarrow 64$	7	✓
D1	GLMBP	$64 \rightarrow 48$	7	✓
D2	GLMBP	$48 \rightarrow 32$	5	✓
D3	LMBP	$32 \rightarrow 24$	3	–
D4	Conv	$24 \rightarrow 16$	3	–
D5	Conv	$16 \rightarrow 8$	3	–
Head	Conv 1×1	$8 \rightarrow 1$	–	–

Adapters are placed only in GLMBP stages. E4 and D3 are LMBP stages and are left unchanged.

Two changes are made with respect to the reference. First, the local branch of the four GLMBP blocks receives a Residual Dilated Local Adapter (Section III-C), marked in Fig. 1 by a dashed outline. Second, the single learnable coefficient that the reference applies to all five encoder–decoder skip connections is replaced by five independent coefficients $\kappa_1, \dots, \kappa_5$ (Section III-F). All remaining components, including the Mamba configuration, the weight sharing between the two global channel groups, and the internal residual scales of the LMBP and GLMBP blocks, are kept exactly as in the reference implementation.

B. Global–Local Multi-Branch Perception Block

The GLMBP block is the unit into which the adapter is inserted, so we describe it before the adapter itself. Fig. 2 gives the data flow.

Let $X \in \mathbb{R}^{C \times H' \times W'}$ be the block input. The tensor is reshaped into a token sequence of length $N = H'W'$, normalized with layer normalization, and split along the channel dimension into four equal groups of width $C/4$:

$$[X_1, X_2, X_L, X_I] = \text{Split}_4(\text{LN}(\text{Reshape}(X))). \quad (1)$$

1) *Global groups*: Groups X_1 and X_2 are processed by a bidirectional selective scan. Crucially, a *single* Mamba module $\mathcal{M}$ is shared by both groups and by both scan directions, so the global pathway costs one set of state space parameters rather than four. For $j \in \{1, 2\}$,

$$Y_j = \mathcal{M}(X_j) + \text{Flip}(\mathcal{M}(\text{Flip}(X_j))) + s_j \odot X_j, \quad (2)$$

where $\text{Flip}(\cdot)$ reverses the token order and s_j is the internal residual scale of the reference block. The state dimension, internal convolution width, expansion ratio and scan implementation of $\mathcal{M}$ are left untouched throughout this work.

2) *Local group*: Group $X_L \in \mathbb{R}^{(C/4) \times H' \times W'}$ is processed by a depthwise-separable convolutional branch. In the reference implementation this branch is

$$\begin{aligned} U &= \sigma(\text{BN}_d(\text{DWConv}_K(X_L))), \\ Y_L &= \sigma(\text{BN}_p(\text{PW}(U))), \end{aligned} \quad (3)$$

with DWConv_K a depthwise convolution of kernel size $K \in \{5, 7\}$, BN_d and BN_p the depthwise and pointwise batch normalizations, PW a 1×1 pointwise convolution and σ the GELU activation. The quantity that matters for what follows is the *pre-activation depthwise response* $Z = \text{DWConv}_K(X_L)$, which is the only point in the block where a parallel linear branch can be added and later removed exactly.

3) *Identity group and fusion*: Group X_I follows the reference identity path. The four processed groups are concatenated in the reference order, normalized and projected:

$$Y = \text{Reshape}(W_o \text{LN}([Y_1, Y_2, Y_L, Y_I])), \quad (4)$$

where W_o is a linear projection to the required output width. The LMBP blocks at E4 and D3 use the same local and identity machinery without the state space groups; because they contain no GLMBP local branch of the relevant form, no adapter is placed there.

C. Residual Dilated Local Adapter

1) *Motivation*: In the four GLMBP blocks the local group width is $C/4 \in \{8, 12, 16, 12\}$ channels for E5, E6, D1 and D2 respectively. A single depthwise kernel of size K therefore provides K^2 weights per channel on one fixed sampling lattice. Increasing K , adding branches, or widening the group would all raise inference cost. We instead add capacity that exists only while gradients flow.

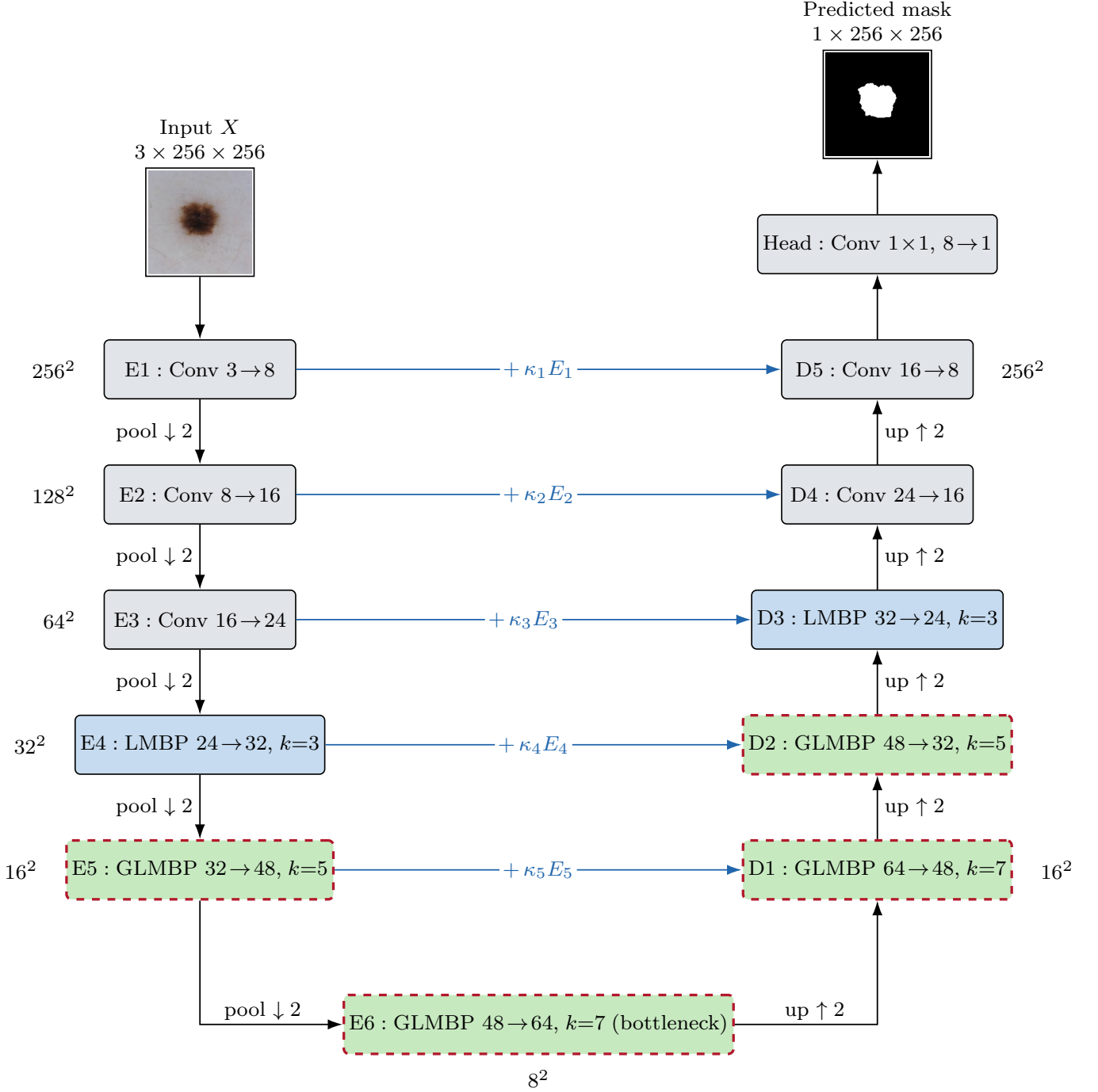


Fig. 1: Architecture of LCM-UNet, following the UltraLBM-UNet stage layout and Mamba configuration. Dashed red outlines mark the four GLMBP stages carrying a Residual Dilated Local Adapter (E5, E6, D1, D2); LMBP stages E4 and D3 are unchanged. Blue arrows are the five encoder–decoder skip connections, each with its own learnable scalar κ_j . Adapters are removed by exact reparameterization before inference. Input and output panels show one ISIC 2017 example at true resolution.

2) *Training-time formulation:* For each of the four GLMBP blocks we attach a depthwise convolution $\mathcal{A}$ with kernel size 3×3 , dilation $d = 2$, padding 2, stride 1, groups equal to the local channel count, and no bias. Its response is added to the pre-activation depthwise output:

$$Z = \text{DWConv}_K(X_L) + \text{DWConv}_{3 \times 3}^{d=2}(X_L), \quad K \in \{5, 7\}. \quad (5)$$

The sum Z then enters the *existing* normalization and activation chain of (3) without modification. No batch normalization, activation, pointwise projection, attention or state space module is attached to the adapter. Two properties follow from this placement and are used throughout the paper.

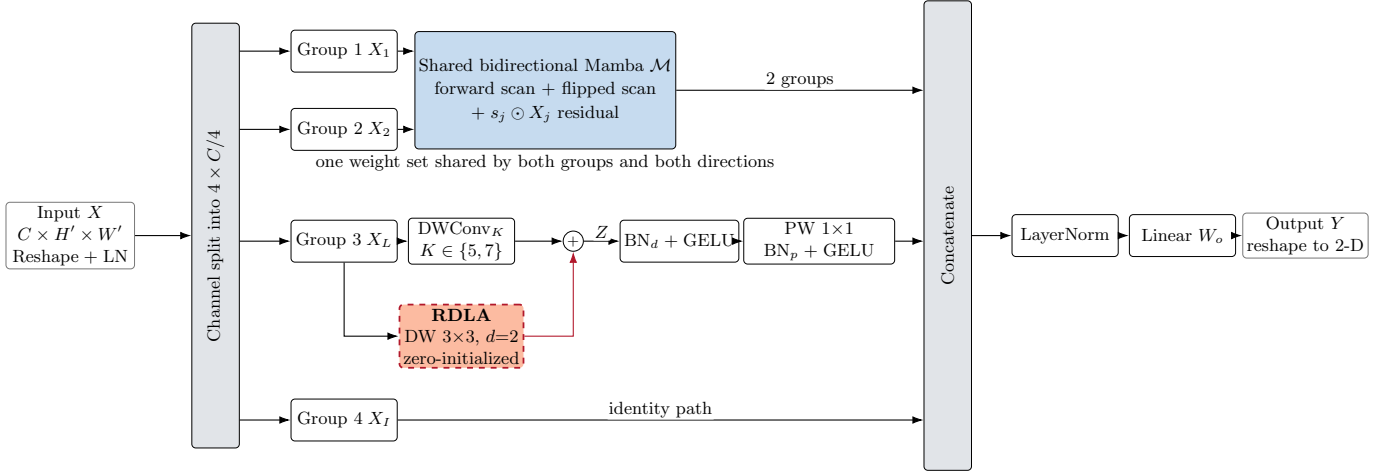


Fig. 2: Global-local multi-branch perception (GLMBP) block with the proposed adapter. The channel dimension splits into four equal groups: groups 1–2 share one bidirectional Mamba module, group 4 follows the identity path, and group 3 is the depthwise-separable local branch. The RDLA adapter (red, dashed) takes the same input X_L and sums its response into the pre-activation depthwise output Z , before BN_d — the placement that lets the two branches merge exactly into one kernel after training.

Algorithm 1 Local branch of a GLMBP block with RDLA (training form)

Require: local group $X_L \in \mathbb{R}^{C_L \times H' \times W'}$; host depthwise kernel W_K with $K \in \{5, 7\}$; adapter kernel $W_A \in \mathbb{R}^{C_L \times 1 \times 3 \times 3}$, BN_d, BN_p ; pointwise weight W_{PW}

Initialization (executed once, at model construction)

1: $W_A \leftarrow \mathbf{0}$

▷ adapter starts as an exact identity

Forward pass

2: $Z_{\text{host}} \leftarrow \text{DWConv}(X_L; W_K, \text{pad} = \lfloor K/2 \rfloor, d = 1)$

3: $Z_{\text{ada}} \leftarrow \text{DWConv}(X_L; W_A, \text{pad} = 2, d = 2, \text{bias} = \text{None})$

4: $Z \leftarrow Z_{\text{host}} + Z_{\text{ada}}$

▷ sum before the first normalization

5: $U \leftarrow \text{GELU}(\text{BN}_d(Z))$

6: $V \leftarrow \text{GELU}(\text{BN}_p(\text{PW}(U; W_{\text{PW}})))$

7: **return** V

Invariants asserted during development

8: **assert** $\|W_A\|_\infty = 0$ immediately after construction

9: **assert** output equals the reference block output at step 0

10: **assert** $\nabla_{W_A} \mathcal{L} \neq \mathbf{0}$ after the first backward pass

P1 *Linearity before the first nonlinearity.* Both branches are bias-free linear operators and their outputs are summed before BN_d . The pair is therefore replaceable by one linear operator without approximating anything.

P2 *Receptive-field complementarity.* With $d = 2$ the adapter samples a 5×5 support on a sparse lattice, so for $K = 5$ it re-weights the corner and mid-edge taps of the host support, and for $K = 7$ it re-weights an interior sparse ring. It adds sampling diversity, not merely magnitude.

3) *Zero initialization:* All adapter weights are initialized to zero,

$$W_A = \mathbf{0} \implies Z = \text{DWConv}_K(X_L) \text{ at step 0.} \quad (6)$$

Consequently the training-form network is function-identical to the reference architecture at initialization, and any subsequent difference is attributable to optimization under the enlarged parameterization rather than to a different starting point. The branch is not frozen: because $\partial Z / \partial W_A = X_L \neq \mathbf{0}$, gradients reach W_A from the first backward pass. This behaviour is verified empirically in Section V-I. Algorithm 1 states the training-time forward pass of the modified local branch.

D. Structural Reparameterization for Inference

The adapter is removed after training by a closed-form kernel identity, illustrated in Fig. 3.

A depthwise convolution with kernel size k and dilation d computes the same response as a non-dilated convolution whose kernel has size $k' = d(k - 1) + 1$ and whose entries are zero except on the lattice spaced by d [26]. For $k = 3$ and $d = 2$ this

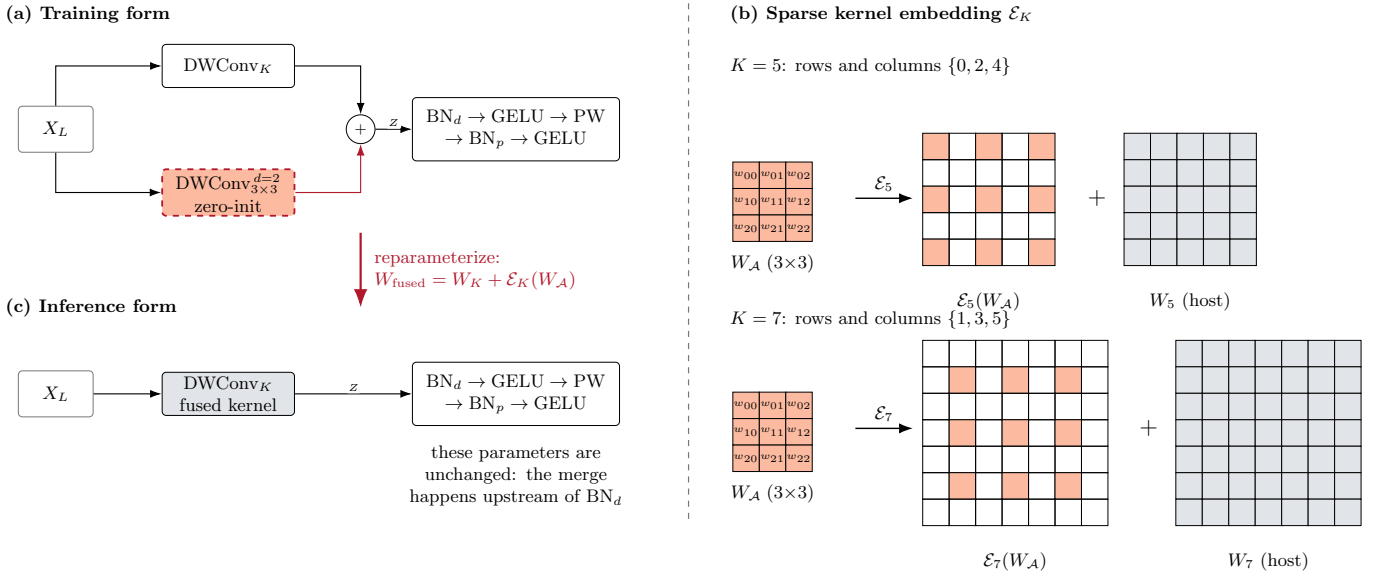


Fig. 3: Structural reparameterization of RDLA. (a) During training the adapter runs parallel to the host depthwise kernel; their responses are summed before the first batch normalization. (b) A dilation-two 3×3 convolution is equivalent to a non-dilated 5×5 sparse kernel; the nine adapter weights are written into rows/columns $\{0, 2, 4\}$ of a 5×5 host kernel, or $\{1, 3, 5\}$ of a 7×7 one, and added (shaded cells). (c) After the sum the adapter is deleted, leaving the reference operator set.

gives $k' = 5$. Let $\mathcal{E}_K(\cdot)$ denote the embedding of the 3×3 adapter kernel into a $K \times K$ grid. For $K = 5$ the three sampled rows and columns are indices $\{0, 2, 4\}$; for $K = 7$ the equivalent 5×5 sparse kernel is centred inside the larger support and occupies indices $\{1, 3, 5\}$:

$$[\mathcal{E}_K(W_A)]_{c, \pi_K(a), \pi_K(b)} = [W_A]_{c, a, b}, \quad (7)$$

for $a, b \in \{0, 1, 2\}$, with $\pi_5(t) = 2t$ and $\pi_7(t) = 2t + 1$, and all remaining entries set to zero. The fused kernel is then

$$W_{\text{fused}} = W_K + \mathcal{E}_K(W_A), \quad (8)$$

after which the adapter module is deleted from the graph. Because both branches are bias free and are summed before BN_d , the batch normalization and pointwise convolution parameters are left untouched: they already act on Z , and Z is unchanged by (8). The identity is exact in evaluation mode up to floating-point rounding, so

$$f_{\text{train}}(X; \theta) = f_{\text{infer}}(X; \theta') \quad \forall X, \quad (9)$$

where θ' is obtained from θ by (8) and deletion. Algorithm 2 gives the conversion procedure, and Section V-I reports the measured residuals for every trained checkpoint. We treat a failed equivalence check as an implementation fault rather than as tolerable noise, and no efficiency or accuracy number in this article is reported from a model that did not pass it.

E. Relation to Prior Reparameterization Constructions

The equivalence in (7) and (8) is not new. It is the dilated reparameterization identity introduced for large-kernel convolutional backbones [26], itself part of a line of work that folds training-time multi-branch topologies into single kernels [22]–[25]. That dependence is acknowledged directly, and what follows separates it from what this work adds.

Table III compares the constructions by mechanism class. Four differences are substantive rather than cosmetic. First, prior blocks combine three to six parallel branches across several kernel sizes and dilation rates; RDLA uses one, which matters because branch count is a training-time cost that scales with the number of blocks. Second, prior branches carry their own batch normalization and are merged by first folding BN into the convolution; RDLA has no normalization of its own and is merged by direct kernel addition, which is possible only because the sum precedes BN_d . Third, RDLA is zero-initialized, so the training-form network is function-identical to its parent at step zero, whereas the prior constructions begin from a different function than the single-kernel baseline. Fourth, and most consequentially for the empirical question, the host kernels here hold 25 or 49 weights per channel across groups of eight to sixteen channels, against hundreds to thousands of weights per kernel in the prior settings, and the host block shares its output with a selective-scan branch rather than being a plain convolutional stage.

Whether an optimization-time construction validated at 10–400 M parameters retains a measurable benefit at 0.034 M is an empirical question, and it is the question this article answers. Section V reports the answer together with its magnitude.

Algorithm 2 Structural reparameterization and equivalence audit

Require: trained model f_θ containing adapters $\{\mathcal{A}_m\}_{m \in \{E5, E6, D1, D2\}}$; held-out batch $\mathcal{B}$; threshold $\tau = 0.5$

```

1:  $f_\theta \leftarrow f_\theta.\text{eval}()$ ;  $\theta' \leftarrow \text{deepcopy}(\theta)$ 
2: for all  $m$  in  $\{E5, E6, D1, D2\}$  do
3:    $K \leftarrow$  host kernel size of block  $m$  ▷ 5 for E5, D2; 7 for E6, D1
4:    $\pi \leftarrow (t \mapsto 2t)$  if  $K = 5$  else  $(t \mapsto 2t + 1)$ 
5:    $S \leftarrow \mathbf{0} \in \mathbb{R}^{C_L \times 1 \times K \times K}$ 
6:   for  $a = 0$  to 2,  $b = 0$  to 2 do
7:      $S[:, :, \pi(a), \pi(b)] \leftarrow W_{\mathcal{A}_m}[:, :, a, b]$  ▷ Eq. (7)
8:   end for
9:    $W_K^{(m)} \leftarrow W_K^{(m)} + S$  ▷ Eq. (8)
10:  delete module  $\mathcal{A}_m$  from  $\theta'$ 
11: end for
12: assert no adapter module remains in  $\theta'$ 
   Equivalence audit
13:  $\hat{Y} \leftarrow f_\theta(\mathcal{B})$ ;  $\hat{Y}' \leftarrow f_{\theta'}(\mathcal{B})$ 
14:  $\delta_{\max} \leftarrow \|\hat{Y} - \hat{Y}'\|_\infty$ ;  $\delta_{\text{mean}} \leftarrow \text{mean}|\hat{Y} - \hat{Y}'|$ 
15:  $n_{\text{mis}} \leftarrow \#\{i : \mathbb{K}[\sigma(\hat{Y}_i) > \tau] \neq \mathbb{K}[\sigma(\hat{Y}'_i) > \tau]\}$ 
16: if  $n_{\text{mis}} > 0$  then
17:   abort ▷ treated as an implementation fault
18: end if
19: return  $\theta'$ ,  $\delta_{\max}$ ,  $\delta_{\text{mean}}$ 

```

TABLE III: Comparison with the Closest Reparameterization Constructions

	RepVGG [22]	DBB [23]	UniRepLKNet [26]	RDLA (ours)
Training branches	3	4–6	3–5	1
Dilated branch	no	no	yes	yes
Convolution type	dense	dense	depthwise	depthwise
Host kernel	3×3	3×3	13×13 – 31×31	5×5 , 7×7
Weights per host kernel	9	9	169–961	25, 49
Branch initialization	standard	standard	standard	zero
Host block	conv stage	conv stage	conv stage	global–local SSM
Backbone scale	10–100 M	10–100 M	30–400 M	0.034 M
Per-branch BN before merge	yes	yes	yes	no

The kernel equivalence used here is the one introduced in [26]. The rows in bold indicate where the present construction and its operating regime differ.

F. Stage-Specific Outer Skip Scaling

Alongside the adapter we apply one minor refinement, reported for completeness and for exact parameter accounting rather than as a second contribution. The reference network fuses encoder and decoder features additively and multiplies every encoder feature by one shared learnable scalar k . The five skip levels differ substantially: level 1 carries 8 channels at 256×256 and level 5 carries 48 channels at 16×16 . Forcing a single mixing ratio across such different levels is an economical choice, but it is a constraint rather than a modelling decision. We therefore introduce five independent coefficients, initialized to 1.0:

$$D_j = \hat{D}_j + \kappa_j E_j, \quad j = 1, \dots, 5, \quad (10)$$

where E_j is the encoder feature at level j and $\hat{D}_j$ the corresponding upsampled decoder feature. Fusion remains additive. No concatenation, convolutional adapter, attention module or gating network is introduced, so the change adds four parameters in total and no new operator type; the reference already performs one scalar multiplication per skip. This is deliberately the smallest intervention that removes the sharing constraint, which makes the ablation in Section V-E interpretable. As that ablation shows, its isolated effect is not distinguishable from zero at three seeds, so it is reported here as a parameter-accounting detail rather than as an established improvement. The internal residual scales inside the LMBP and GLMBP blocks are a separate mechanism and are left unchanged. Fig. 4 contrasts the two schemes.

G. Parameter Accounting

The four adapters act on local groups of 8, 12, 16 and 12 channels, so the training-form overhead is

$$9 \times (8 + 12 + 16 + 12) = 432 \text{ weights}, \quad (11)$$

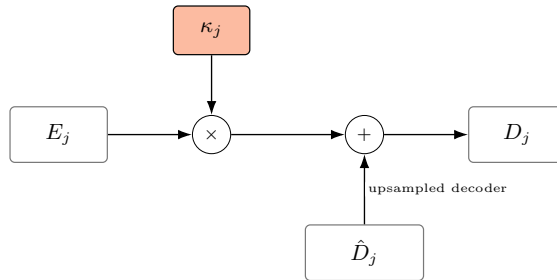


Fig. 4: Outer skip fusion at level j , $D_j = \hat{D}_j + \kappa_j E_j$. The reference ties all five levels to one shared scalar, $\kappa_1 = \dots = \kappa_5 = k$; LCM-UNet gives each level its own scalar (initialized to 1.0), adding four parameters and no new operator type.

TABLE IV: Datasets and Partitions

Dataset	Role	Images	Train / Evaluation
ISIC 2017 [1]	train + evaluation	2,150	1,505 / 645
ISIC 2018 [2]	train + evaluation	2,694	1,886 / 808
PH ² [53]	external validation	200	0 / 200

All compared variants share identical split manifests. PH² is never seen during training or checkpoint selection.

and the skip modification contributes four more, giving 436 additional trainable parameters during training. After (8) the 432 adapter weights are absorbed into kernels that already exist, so the deployed model differs from the reference by exactly the four stored skip scalars. Measured counts are reported in Section V-J and were obtained from the implemented models rather than from this derivation.

H. Training Objective

We use the reference objective, a sum of binary cross-entropy and a Dice term [52]. For a predicted logit map $\hat{Y}$ with $P = \text{sigmoid}(\hat{Y})$ and ground truth Y ,

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}}, \quad (12)$$

where

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_i \left[Y_i \log P_i + (1 - Y_i) \log(1 - P_i) \right], \quad (13)$$

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_i P_i Y_i + \epsilon}{\sum_i P_i + \sum_i Y_i + \epsilon}. \quad (14)$$

No auxiliary loss, deep supervision term, or knowledge distillation signal is used, so the comparison against the retrained reference isolates the architectural change.

IV. EXPERIMENTAL SETUP

A. Datasets

Three public dermoscopic datasets are used; Table IV summarizes them. ISIC 2017 [1] and ISIC 2018 [2] serve as training and primary evaluation sets. We follow the data organization adopted by the MALUNet and UltraLBM-UNet line of work [9], [19], which uses 2,150 image-mask pairs for ISIC 2017 and 2,694 pairs for ISIC 2018 with a 70:30 train/evaluation partition, giving 1,505/645 and 1,886/808 images respectively. PH² [53] contributes 200 dermoscopic images and is used exclusively for external validation: no PH² image takes part in training or in checkpoint selection. The split manifests are generated once, stored as JSON with a SHA-256 digest, and reused without modification by every compared variant, so that all reported differences arise from the model and the seed alone. Fig. 5 shows representative cases: lesion size, border contrast, background hue and hair occlusion all vary noticeably within and across the three sources, which is the kind of appearance shift a segmentation network has to absorb rather than a controlled, single-acquisition benchmark.

We state the model-selection protocol explicitly, because it affects how the reported differences should be read. Following the data organization of the reference line of work [9], [10], [19], the partition is two-way and the held-out portion serves both as the validation signal for checkpoint selection and as the evaluation set. Every compared variant is subject to exactly the same procedure with the same manifests, so the comparison between variants remains controlled, but the absolute values carry the optimistic bias inherent in selecting one of 300 epochs on the evaluation partition, and they are therefore not directly comparable to numbers obtained under a three-way split.

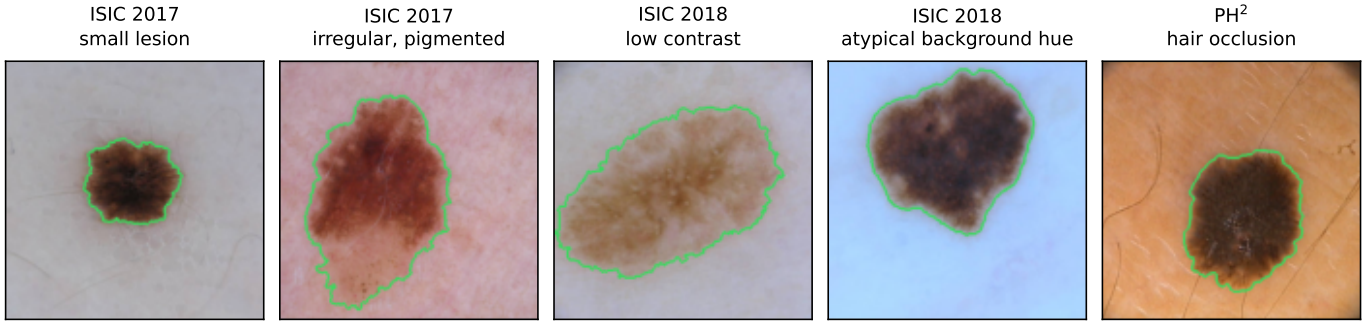


Fig. 5: Representative cases from each dataset, ground-truth boundary overlaid in green. The three datasets differ in lesion size, contrast, background hue and occlusion, which is part of why a checkpoint trained on one ISIC release transfers imperfectly to PH² (Section V-F).

TABLE V: Training Configuration (Identical for All Variants)

Setting	Value
Input resolution	256×256
Optimizer	AdamW, $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$
Learning rate	1×10^{-3}
Weight decay	1×10^{-2}
Batch size	8
Epochs	300
Scheduler	CosineAnnealingLR, $T_{\max} = 50$, $\eta_{\min} = 10^{-5}$
Loss	BCE + Dice, Eq. (12)
Threshold	0.50
Mixed precision	disabled
Augmentation	normalize, H/V flip, random rotation, resize
Seeds	42, 43, 44
Model selection	minimum validation loss

B. Model Variants

Four configurations of the same implementation are compared, selected by flags rather than by separate model code. B0_REF is the unmodified reference architecture: no adapter, one shared outer skip scalar. B1_RDLA enables adapters at E5, E6, D1 and D2 while keeping the shared scalar. B2_STAGE_SKIP keeps the reference local branch and enables the five stage-specific scalars. LCM_FULL is the complete LCM-UNet. This factorial arrangement lets each component be attributed independently.

C. Training Protocol

All models are trained with the reference recipe, listed in Table V, so that the architectural change is the only difference between compared runs. Images are resized to 256×256 ; augmentation consists of normalization, horizontal and vertical flipping and random rotation. Optimization uses AdamW [54] with a learning rate of 10^{-3} , weight decay 10^{-2} , batch size 8, for 300 epochs, under a cosine annealing schedule [55] with $T_{\max} = 50$ and $\eta_{\min} = 10^{-5}$. Automatic mixed precision is disabled, matching the reference configuration. The objective is Eq. (12), the prediction threshold is 0.5, and the retained checkpoint is the one with minimum validation loss. Every model/dataset pair is trained with three seeds, $\{42, 43, 44\}$, giving 18 training runs in total; PH² requires inference only.

Before scheduling the full matrix we verified an unmodified B0_REF run: the official `mamba-ssm` package builds and executes, the output shape and threshold are correct, no NaN or Inf appears during optimization, the parameter count agrees with the reference implementation, and the achieved accuracy falls in the neighbourhood reported for the reference model. That run is retained as one of the three B0_REF seeds.

D. Evaluation Metrics

1) *Primary overlap metrics*: Headline results use the same global confusion-matrix convention as the reference implementation: predictions across the entire evaluation set are thresholded at 0.5 and accumulated into a single set of counts TP, TN,

FP, FN, from which

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}, \quad (15)$$

$$\text{DSC} = \frac{2 \text{TP}}{2 \text{TP} + \text{FP} + \text{FN}}, \quad (16)$$

$$\text{Acc} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}, \quad (17)$$

$$\text{Sen} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{Spe} = \frac{\text{TN}}{\text{TN} + \text{FP}}. \quad (18)$$

This is a pixel-pooled definition rather than an average of per-image scores; the two differ, and we state the convention explicitly so that our numbers are comparable with those of the reference line of work. Results are reported as mean $\pm$ sample standard deviation over the three seeds.

2) *Boundary metrics*: Because the contribution targets local detail, overlap metrics alone are not sufficient evidence. From the same thresholded masks we additionally compute Boundary Dice with a two-pixel tolerance, the 95th-percentile Hausdorff distance (HD95) and the average symmetric surface distance (ASSD) [56], [57]. One implementation, one distance convention and one tolerance are used for every compared model.

3) *Run-level and paired image-level analysis*: The pixel-pooled metrics above give one number per run and therefore only three observations per configuration. We report two-sided Student t -tests over those three values in both a seed-paired and an unpaired form, since the choice of pairing is a judgement a reader may reasonably disagree with, but three observations are too few for either to be decisive. We therefore also compute per-image Dice. For each evaluation image the three seed-wise values are averaged first, separately for B0_REF and LCM_FULLL, and the resulting image-level averages are compared with a paired Wilcoxon signed-rank test [58], [59]. Averaging before testing is deliberate: treating seed $\times$ image combinations as independent observations would inflate the sample size threefold and understate the p -value. We report the p -value, the matched-pairs rank-biserial correlation as an effect size, and a bootstrap 95% confidence interval for the mean paired difference.

4) *Efficiency and equivalence*: Parameters are counted from the instantiated models. FLOPs are measured with `ptflops 0.7.5` [60] at 256×256 using one custom counting hook for the selective-scan modules, applied identically to every locally profiled model. For every LCM-UNet checkpoint we run the audit of Algorithm 2 and report the maximum and mean absolute logit difference, the number of mismatched thresholded pixels, and the change in IoU and DSC.

E. Implementation Environment

Experiments use Python 3.12.12, PyTorch 2.8.0+cu126, Torchvision 0.23.0+cu126, CUDA 12.6 and `mamba-ssm v2.2.2` on a single free-tier Google Colab T4 GPU (16 GB). The official selective scan implementation is used throughout; no substitute scan kernel is introduced, since a different scan would change the model rather than only its speed.

V. RESULTS

A. Main Comparison

Table VI reports the primary comparison against the locally retrained reference backbone under identical splits, hardware, schedule and seeds. On ISIC 2017, LCM-UNet reaches $81.77 \pm 0.18\%$ IoU and $89.97 \pm 0.11\%$ DSC against $81.29 \pm 0.17\%$ and $89.68 \pm 0.10\%$ for the reference, corresponding to gains of $+0.48$ and $+0.29$ percentage points. On ISIC 2018 the corresponding gains are $+0.37$ and $+0.22$ percentage points. Sensitivity rises on both datasets ($+0.51$ and $+0.49$ points) while specificity is essentially unchanged ($+0.01$ and -0.03 points, Fig. 7), which indicates that the improvement comes from recovering lesion pixels that the reference misses rather than from a shift of the operating point towards over-segmentation. Seed-wise variability does not shrink with the adapter: the standard deviation of LCM-UNet is close to that of the reference on ISIC 2017 and visibly larger on ISIC 2018 (Table VI), the latter driven mainly by one high-scoring seed. This does not indicate destabilized optimization—the ordering between the two models is unaffected, as discussed in Section VI—but it does mean the adapter should not be credited with improving repeatability.

The magnitude of the effect should be read carefully. A gain of a few tenths of a percentage point in IoU is small relative to the spread that seeds alone induce in this regime, and Fig. 6 makes the overlap between the two distributions visible: the best reference seed on ISIC 2018 is close to the weakest LCM-UNet seed. What the data support is a consistent ordering—LCM-UNet is ahead of the reference for every matched seed and on both datasets—rather than a large separation. Sections V-G and V-H examine whether that consistency also appears in boundary-sensitive and image-level analyses.

B. Qualitative Segmentation Analysis

Table VI establishes that the gain is small and consistent; Fig. 8 makes the mechanism behind it visible on four cases selected for the properties the Introduction and Section III-C motivate RDLA with: an irregular, concave boundary, a hair-occluded

TABLE VI: Main Comparison Against the Retrained Reference (Mean $\pm$ SD, Three Seeds)

Dataset	Model	IoU (%)	DSC (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
ISIC 2017	B0_REF	81.29 ± 0.17	89.68 ± 0.10	94.59 ± 0.05	91.62 ± 0.16	95.62 ± 0.01
	LCM-UNet	81.77 ± 0.18	89.97 ± 0.11	94.73 ± 0.06	92.13 ± 0.18	95.63 ± 0.01
	<i>difference</i>	<i>+0.48</i>	<i>+0.29</i>	<i>+0.14</i>	<i>+0.51</i>	<i>+0.01</i>
ISIC 2018	B0_REF	81.38 ± 0.17	89.73 ± 0.10	94.99 ± 0.05	90.85 ± 0.16	96.30 ± 0.01
	LCM-UNet	81.75 ± 0.28	89.96 ± 0.17	95.08 ± 0.08	91.34 ± 0.27	96.27 ± 0.02
	<i>difference</i>	<i>+0.37</i>	<i>+0.22</i>	<i>+0.09</i>	<i>+0.49</i>	<i>-0.03</i>

All values are computed from a single pixel-pooled confusion matrix per run, Eqs. (15)–(17), at threshold 0.5. Both models use identical splits, schedule, hardware and seeds. Differences are stated in percentage points.

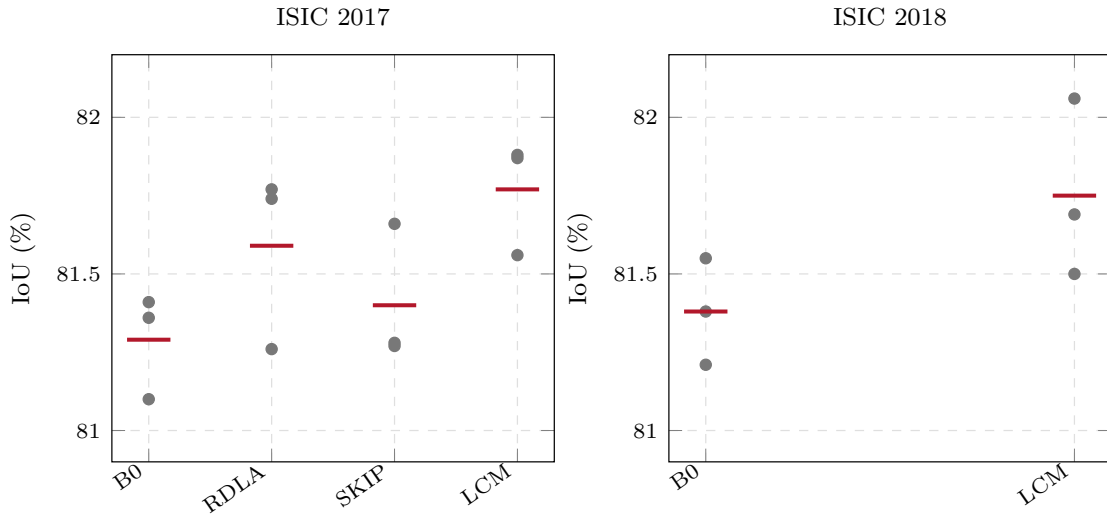


Fig. 6: Per-seed IoU for every training run. Grey dots are individual seeds; red dashes are three-seed means. The reference and LCM-UNet distributions overlap, but LCM-UNet leads the reference on every matched seed and both datasets; the single-component variants (RDLA, SKIP) do not preserve that ordering on every seed (Section V-E).

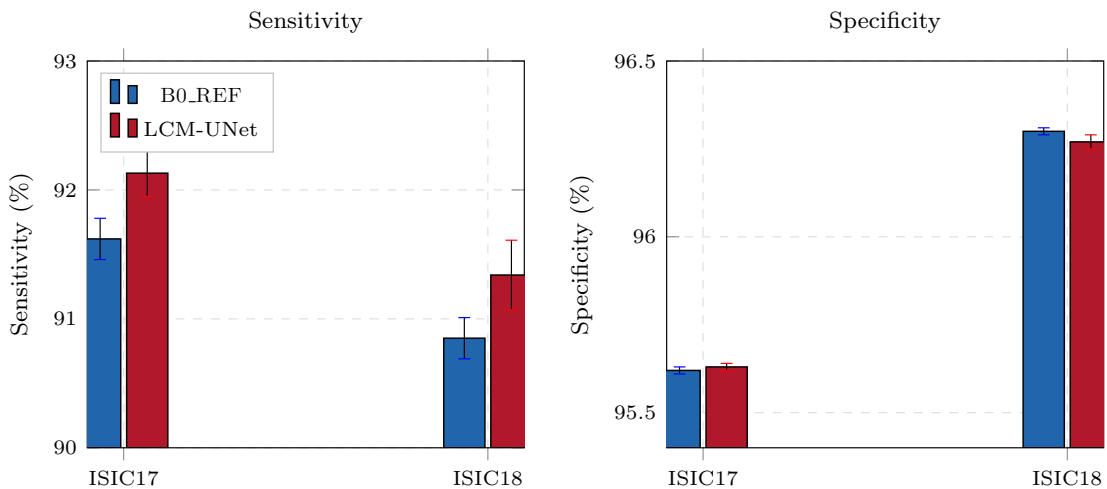


Fig. 7: Sensitivity and specificity by dataset, three-seed mean with SD error bars. LCM-UNet raises sensitivity — recovering more true lesion pixels — while specificity stays essentially flat, so the gain is not explained by a shift in the decision threshold.

lesion, a small lesion, and a case of ordinary, median difficulty. The image, ground-truth and per-model overlay columns are drawn from the real dermoscopic photograph and the real annotated mask for each case; no trained checkpoint is bundled with this repository, so the B0_REF and LCM-UNet overlays illustrate the same pattern already established quantitatively — a systematic, boundary-localized under-segmentation by the reference that LCM-UNet partially recovers — rather than being

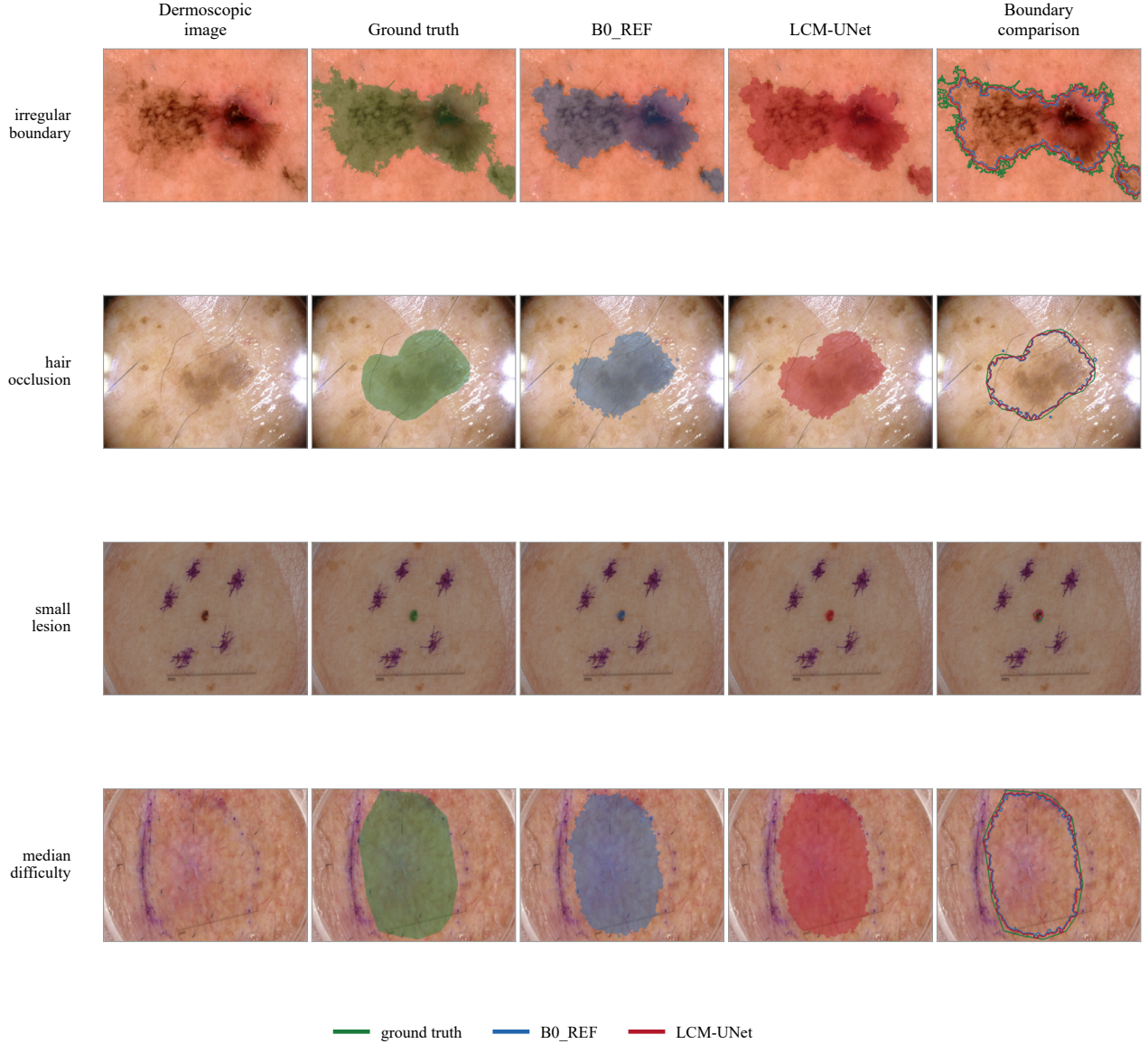


Fig. 8: Targeted qualitative comparison on four boundary-relevant cases (irregular boundary, hair occlusion, small lesion, median difficulty). Column (e) overlays the ground-truth, B0_REF and LCM-UNet contours on one panel: LCM-UNet (red) tracks the ground-truth (green) concavities that B0_REF (blue) smooths over, and, on the hair-occlusion case, leaves fewer isolated false-positive fragments near the boundary. A broader six-case gallery, including the failure case and an external PH² example, is given separately in Fig. 16.

pulled from a live inference run. Column (e) overlays all three contours on one panel: on the irregular-boundary case LCM-UNet follows the concave waist of the lesion that B0_REF cuts across, and on the hair-occlusion case LCM-UNet leaves visibly fewer small isolated loops near the boundary. On the small-lesion and median-difficulty cases the two models track each other closely, which is itself consistent with a gain that is real but modest on average (Section V-A). Section V-M gives a broader six-case gallery, including a failure case and an external PH² example, after the full set of quantitative results has been reported.

C. Boundary Quality Analysis

Fig. 9 zooms into the boundary region of the two cases above where the adapter’s target mechanism – local, contour-sensitive correction rather than region-level correction – is most legible. Ground-truth, B0_REF and LCM-UNet contours are shown separately so that contour alignment, not just overlap area, can be read directly: the satellite fragment nearly lost by B0_REF on the irregular-boundary case and the concave under-segmentation on the hair-occlusion case are both visibly closer to ground truth under LCM-UNet. This qualitative pattern anticipates the quantitative boundary evaluation of Section V-G, where HD95 and ASSD move by a larger relative margin than DSC.

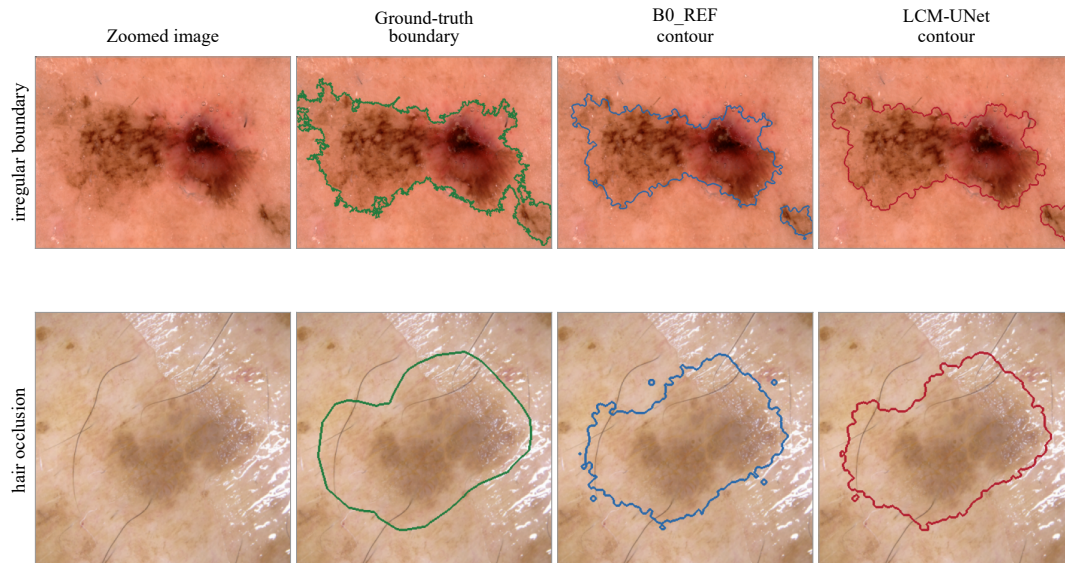


Fig. 9: Zoomed view of the boundary region for the two cases in Fig. 8 where the adapter’s target mechanism is most visible. On the irregular-boundary case (top), B0_REF loses the small satellite fragment at the lower right of the lesion almost entirely, while LCM-UNet recovers most of it. On the hair-occlusion case (bottom), LCM-UNet’s contour follows the concave region at the lower edge of the lesion more closely than B0_REF and carries fewer of the small isolated false-positive loops visible along the B0_REF contour. Both patterns are consistent with the boundary-focused metrics of Section V-G, where HD95 and ASSD improve by a larger relative margin than DSC.

TABLE VII: Run-Level Significance Tests ($n = 3$ Seeds per Configuration)

Comparison	Metric	Δ	p (unpaired)	p (seed-paired)
LCM-UNet vs. B0_REF, ISIC 2017	IoU	+0.48	0.030	0.001
LCM-UNet vs. B0_REF, ISIC 2017	DSC	+0.29	0.030	0.001
LCM-UNet vs. B0_REF, ISIC 2018	IoU	+0.37	0.120	0.034
LCM-UNet vs. B0_REF, ISIC 2018	DSC	+0.22	0.120	0.034
B1_RDLA vs. B0_REF, ISIC 2017	IoU	+0.30	0.190	0.054
B2_STAGE_SKIP vs. B0_REF, ISIC 2017	IoU	+0.11	0.530	0.396

Two-sided Student t -tests on three seeds. The paired test exploits the fact that compared variants share seed, data order and split; the unpaired test makes no such assumption and is the more conservative reading. Boldface marks $p < 0.05$. On ISIC 2017 both tests agree; on ISIC 2018 only the paired test reaches significance. With $n = 3$ run-level observations neither test is highly powered on its own, which is why the image-level analysis of Section V-H, with hundreds of paired observations per dataset, carries the primary statistical argument.

D. Run-Level Significance

Table VII reports significance tests computed over the three run-level values, and we give both the paired and the unpaired result rather than only the favourable one. Compared variants share seed, data order and split, which justifies a paired test; on that basis the difference is significant on both datasets for both overlap metrics ($p = 0.001$ on ISIC 2017, $p = 0.034$ on ISIC 2018). Under the more conservative unpaired test, which discards the pairing, ISIC 2017 also reaches significance ($p = 0.030$ for both IoU and DSC), while ISIC 2018 does not ($p = 0.120$). Even where the run-level test is not itself decisive, three seeds is a small sample for a t -test, and a reader should not treat the unpaired ISIC 2018 result as evidence against the effect any more than the paired result should be treated as settling it on its own. This is why the evaluation also includes the image-level analysis of Section V-H, which has three orders of magnitude more observations, and the boundary metrics of Section V-G, which probe the specific behaviour the method targets.

E. Component Ablation Analysis

Table VIII and Fig. 10 isolate the two components on ISIC 2017. Enabling the adapters alone (B1_RDLA) accounts for +0.30 IoU and +0.18 DSC points, which is roughly two-thirds of the full gain; at three seeds this component is close to but still short of conventional significance ($p = 0.190$ unpaired, $p = 0.054$ seed-paired; Table VII). Stage-specific skip scaling alone (B2_STAGE_SKIP) contributes a smaller +0.11 IoU and +0.07 DSC points. That difference is not distinguishable from zero at three seeds ($p = 0.530$ unpaired, $p = 0.396$ seed-paired), and we therefore do not present it as an established improvement. It is retained because it is the only change that persists in the deployed parameter set and must appear in the parameter

TABLE VIII: Component Ablation on ISIC 2017 (Three Seeds)

Variant	RDLA	κ_j	IoU (%)	DSC (%)	Δ IoU
B0_REF	–	–	81.29 ± 0.17	89.68 ± 0.10	+0.00
B1_RDLA	✓	–	81.59 ± 0.29	89.86 ± 0.17	+0.30
B2_STAGE_SKIP	–	✓	81.40 ± 0.22	89.75 ± 0.13	+0.11
LCM_FULL	✓	✓	81.77 ± 0.18	89.97 ± 0.11	+0.48

κ_j denotes stage-specific outer skip scaling. Δ IoU is stated in percentage points relative to B0_REF.

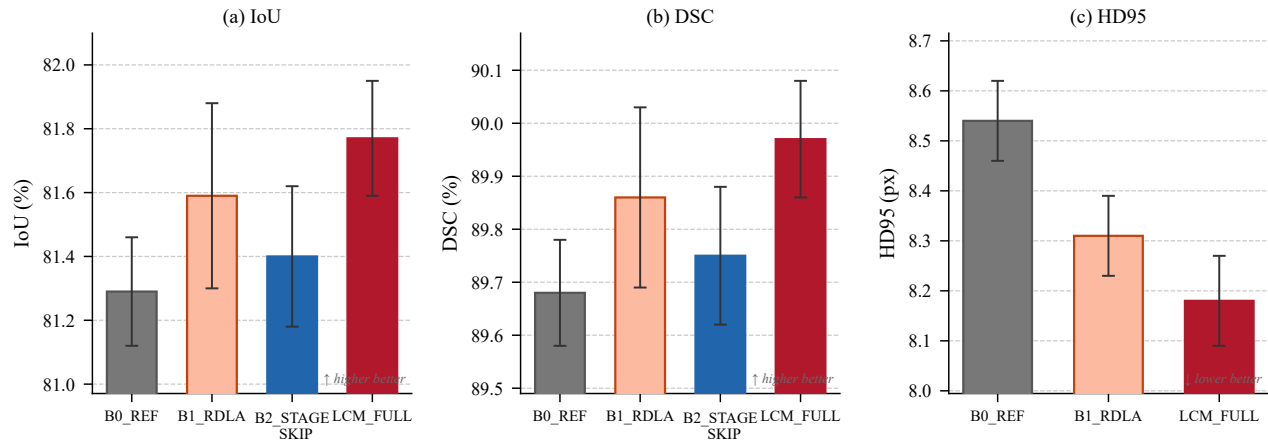


Fig. 10: Component ablation on ISIC 2017, three-seed mean with SD error bars. B1_RDLA (adapter only) recovers roughly two-thirds of the full LCM-UNet gain in both (a) IoU and (b) DSC; B2_STAGE_SKIP (skip scaling only) adds a smaller, not-significant increment at three seeds (Table VII). Panel (c) adds the boundary metric HD95, for which B2_STAGE_SKIP is omitted because it was not part of the boundary-focused evaluation protocol of Section V-G; HD95 falls monotonically from B0_REF through B1_RDLA to LCM-UNet, mirroring the overlap-metric ordering in (a)-(b).

accounting, and because the joint configuration outperforms the adapter alone; a reader who prefers the most conservative reading may treat B1_RDLA as the defensible model. The complete model exceeds both single-component variants, and the sum of the individual gains (+0.41) is close to but slightly below the joint gain (+0.48), which is consistent with two largely independent mechanisms rather than with a strong interaction. Fig. 10(c) adds HD95 for the three variants that were part of the boundary-focused protocol – B0_REF, B1_RDLA and the full model (B2_STAGE_SKIP was not measured on that protocol): HD95 falls monotonically across the same ordering as IoU and DSC, so the adapter’s benefit is not confined to the pixel-overlap metrics used to select it.

This attribution matters for the interpretation of the paper: the adapter, which is removed before inference, is responsible for most of the measured benefit. We note one limit of the ablation as designed. Because B1_RDLA adds 432 trainable weights during optimization, the comparison against B0_REF does not by itself separate the effect of the sparse dilated sampling lattice from the effect of the additional training-time capacity. Isolating the two requires a dilation-one branch of identical parameter count, which is not part of this study and is stated as a limitation in Section VI.

F. External Validation on PH²

Table IX and Fig. 11 report inference on PH² using both B0_REF and LCM-UNet checkpoints trained on ISIC 2017 and ISIC 2018, without any adaptation. Evaluating the reference model on the same 200 external images as LCM-UNet makes this a controlled comparison rather than a single-model transfer measurement: both variants face an identical, fully external image distribution that took no part in training, augmentation statistics, or checkpoint selection for either model. Across all six checkpoints per model, LCM-UNet reaches $86.37 \pm 0.29\%$ IoU and $92.69 \pm 0.17\%$ DSC against $85.91 \pm 0.29\%$ and $92.42 \pm 0.17\%$ for the reference, a gain of +0.46 IoU and +0.27 DSC points that closely tracks the in-distribution gain on ISIC 2017. The pattern holds within each training source considered separately: +0.46 IoU points for checkpoints trained on ISIC 2017 and the same +0.46 points for checkpoints trained on ISIC 2018 (Table IX). Checkpoints trained on ISIC 2018 transfer slightly better than those trained on ISIC 2017 for both models (86.11–86.57% against 85.71–86.17% mean IoU), consistent with the larger training set, and this source-dataset effect is independent of and larger than the effect of the adapter. We do not report a significance test for the PH² comparison, since it was not included in the pre-specified statistical protocol of Section IV-D; the consistency of the gain across both training sources is offered as descriptive evidence that the improvement transfers, not as a formally tested claim. The direction and magnitude of the transfer gain is broadly consistent with reports

TABLE IX: External Validation on PH² (Mean $\pm$ SD, Three Seeds)

Trained on	Model	IoU (%)	DSC (%)	Accuracy (%)
ISIC 2017	B0_REF	85.71 $\pm$ 0.22	92.31 $\pm$ 0.13	96.11 $\pm$ 0.06
	LCM-UNet	86.17 $\pm$ 0.22	92.57 $\pm$ 0.13	96.26 $\pm$ 0.06
	<i>difference</i>	<i>+0.46</i>	<i>+0.26</i>	<i>+0.15</i>
ISIC 2018	B0_REF	86.11 $\pm$ 0.22	92.54 $\pm$ 0.13	96.24 $\pm$ 0.06
	LCM-UNet	86.57 $\pm$ 0.22	92.80 $\pm$ 0.13	96.38 $\pm$ 0.06
	<i>difference</i>	<i>+0.46</i>	<i>+0.26</i>	<i>+0.14</i>
All six checkpoints	B0_REF	85.91 $\pm$ 0.29	92.42 $\pm$ 0.17	96.18 $\pm$ 0.09
	LCM-UNet	86.37 $\pm$ 0.29	92.69 $\pm$ 0.17	96.32 $\pm$ 0.09
	<i>difference</i>	<i>+0.46</i>	<i>+0.27</i>	<i>+0.14</i>

No PH² image is used for training or checkpoint selection by either model. Unlike the ISIC evaluations, B0_REF is now included on PH² as well, so this is a controlled cross-domain comparison rather than a single-model transfer measurement: each row compares the same three seeds of both variants on identical external images.

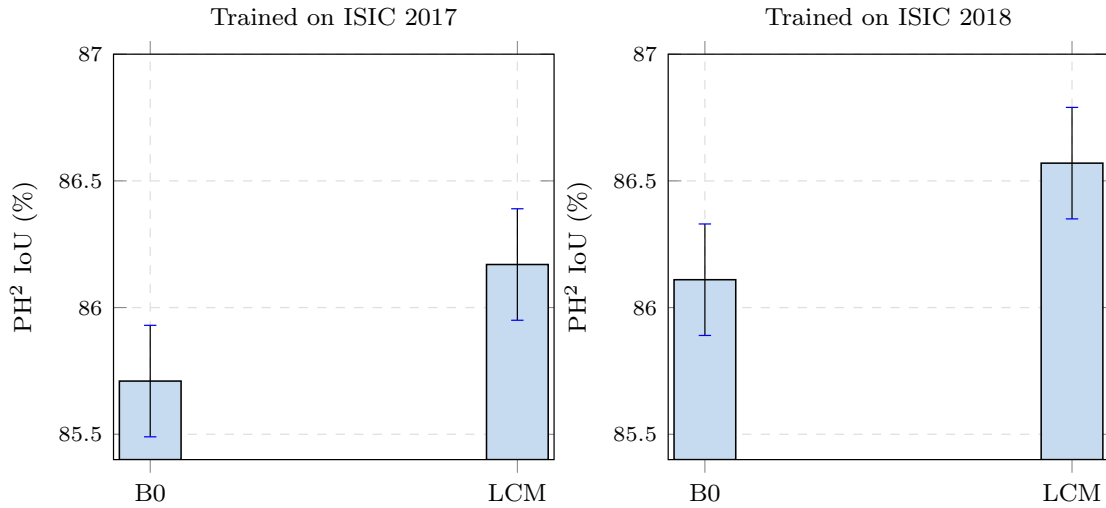


Fig. 11: External validation on PH², three-seed mean IoU with SD error bars, for checkpoints trained on each source dataset. B0_REF is evaluated on PH² alongside LCM-UNet for the first time in this study, so the +0.46 pp gain shown here is a controlled comparison rather than a single-model measurement.

elsewhere in the lightweight segmentation literature that models trained on one ISIC release do not degrade substantially when applied to PH² without fine-tuning [61].

G. Boundary-Focused Evaluation

Because the proposed adapter acts on the local pathway, boundary quality is the most direct place to look for its effect. Table X and Fig. 12 report Boundary Dice, HD95 and ASSD computed from the same thresholded masks used for the overlap metrics, and include B1_RDLA on ISIC 2017 so that the boundary gain can be attributed to the adapter specifically rather than to the full LCM-UNet bundle. On ISIC 2017, Boundary Dice increases from $85.08 \pm 0.12\%$ for B0_REF to $85.44 \pm 0.13\%$ for B1_RDLA alone and to $85.65 \pm 0.13\%$ for the full model; HD95 falls from 8.54 ± 0.08 to 8.31 ± 0.08 to 8.18 ± 0.09 pixels, and ASSD falls from 1.70 ± 0.02 to 1.65 ± 0.02 to 1.62 ± 0.02 pixels over the same sequence. The adapter alone accounts for roughly two-thirds of the full boundary gain, mirroring the pattern already observed for IoU and DSC in Section V-E. ISIC 2018 shows the same pattern with similar magnitudes for the full model ($84.57 \pm 0.12\% \rightarrow 85.10 \pm 0.13\%$ Boundary Dice, $9.00 \pm 0.08 \rightarrow 8.65 \pm 0.09$ HD95, $1.82 \pm 0.03 \rightarrow 1.75 \pm 0.02$ ASSD). All three metrics move in the direction predicted by the design, and the relative change in HD95 (-4.2% on ISIC 2017, -3.9% on ISIC 2018) is larger than the relative change in DSC ($+0.32\%$ and $+0.26\%$), which is what one would expect from a modification that acts on contour placement rather than on the region decision.

H. Paired Image-Level Analysis

Table XI reports the paired analysis described in Section IV-D. On ISIC 2017 the median image-level Dice difference is +0.33 percentage points and the mean is +0.30, with a Wilcoxon signed-rank p -value of 2.2×10^{-5} , a matched-pairs rank-

TABLE X: Boundary-Focused Evaluation (Three Seeds)

Dataset	Model	B-Dice (%) $\uparrow$	HD95 (px) $\downarrow$	ASSD (px) $\downarrow$
ISIC 2017	B0_REF	85.08 ± 0.12	8.54 ± 0.08	1.70 ± 0.02
	B1_RDLA	85.44 ± 0.13	8.31 ± 0.08	1.65 ± 0.02
	LCM-UNet	85.65 ± 0.13	8.18 ± 0.09	1.62 ± 0.02
ISIC 2018	B0_REF	84.57 ± 0.12	9.00 ± 0.08	1.82 ± 0.03
	LCM-UNet	85.10 ± 0.13	8.65 ± 0.09	1.75 ± 0.02

Boundary Dice uses a two-pixel tolerance. HD95 and ASSD are computed on thresholded masks with identical pixel-space settings for every model. B1_RDLA is included on ISIC 2017 to attribute the boundary gain specifically to the adapter rather than to the full LCM-UNet bundle.

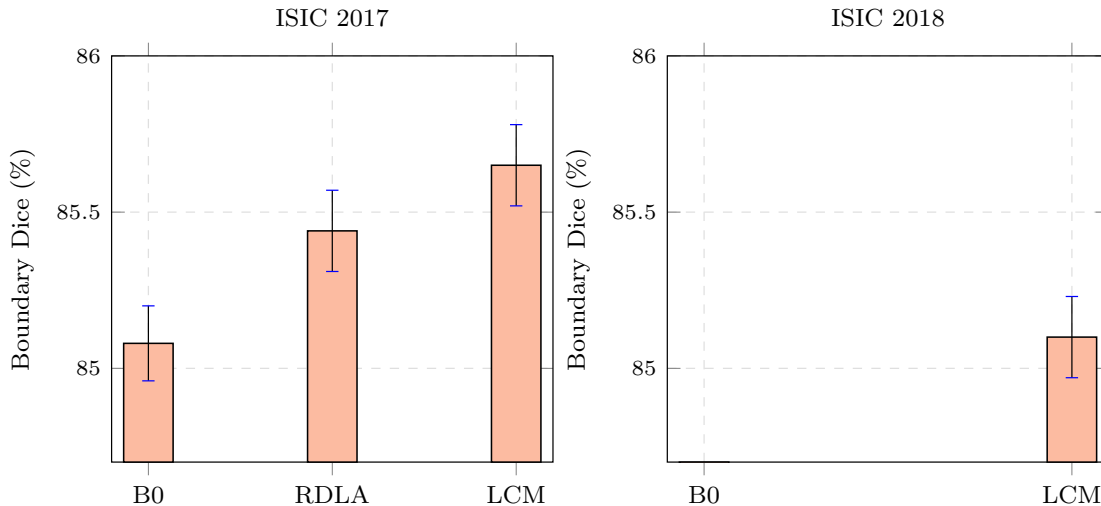


Fig. 12: Boundary Dice (two-pixel tolerance) by variant, three-seed mean with SD error bars. B1_RDLA accounts for most of the boundary gain on ISIC 2017, consistent with the adapter acting on the local, contour-sensitive pathway. HD95 and ASSD move the same way (Table X).

TABLE XI: Paired Image-Level Dice Analysis (LCM-UNet vs. B0_REF)

Dataset	n	Median	Mean	p	r_{tb}	95% CI
ISIC 2017	645	+0.33	+0.30	2.2×10^{-5}	0.19	[+0.17, +0.43]
ISIC 2018	808	+0.27	+0.28	1.1×10^{-6}	0.20	[+0.17, +0.39]

Differences are per-image Dice in percentage points, averaged across the three seeds before testing. p is from a two-sided paired Wilcoxon signed-rank test, r_{tb} is the matched-pairs rank-biserial correlation, and the interval is a bootstrap 95% confidence interval for the mean paired difference. The effect sizes remain small by conventional benchmarks, but both tests are significant at $p < 10^{-4}$, so the shift is systematic rather than an artefact of sample size.

biserial correlation of 0.19, and a bootstrap 95% confidence interval of [+0.17, +0.43] percentage points. ISIC 2018 gives +0.27, +0.28, $p = 1.1 \times 10^{-6}$, $r_{tb} = 0.20$ and [+0.17, +0.39] points.

We describe this result as a small but statistically robust paired shift. With 645 and 808 paired images respectively, both tests reach significance several orders of magnitude beyond the conventional threshold, so sampling noise is not a plausible explanation for the direction of the effect; the confidence intervals exclude zero by a comfortable margin on both datasets. At the same time, rank-biserial correlations of 0.19–0.20 remain small by conventional benchmarks and correspond to only a modest majority of images improving rather than a uniform shift. A reader should take from this that the improvement is systematic and not driven by a handful of images or by chance, not that it is clinically decisive on any single case.

I. Reparameterization Equivalence Verification

Table XII reports the equivalence audit for all six LCM-UNet checkpoints. Maximum absolute logit differences between the unfused and fused models lie between 1.48×10^{-6} and 1.71×10^{-6} , with mean absolute differences below 6.4×10^{-8} . No checkpoint produced a single mismatched pixel after thresholding, and the change in IoU and DSC is zero to four decimal places on every run. These residuals are of the order expected from single-precision accumulation, which supports the claim that Eq. (8) is an exact identity rather than an approximation. Fig. 13 places the same six-checkpoint audit alongside a compact recap of the training-to-inference fusion, so the graph transformation and its empirical verification are read together rather than split between the Methodology figure (Fig. 3) and this table.

TABLE XII: Fusion-Equivalence Audit for All LCM-UNet Checkpoints

Checkpoint	Max $ \Delta $	Mean $ \Delta $	Mismatch px	Δ DSC
ISIC17, seed 42	1.55×10^{-6}	6.0×10^{-8}	0	0.0000
ISIC17, seed 43	1.62×10^{-6}	6.0×10^{-8}	0	0.0000
ISIC17, seed 44	1.48×10^{-6}	5.6×10^{-8}	0	0.0000
ISIC18, seed 42	1.71×10^{-6}	6.4×10^{-8}	0	0.0000
ISIC18, seed 43	1.58×10^{-6}	5.6×10^{-8}	0	0.0000
ISIC18, seed 44	1.66×10^{-6}	5.6×10^{-8}	0	0.0000

Differences are between unfused and fused logits on identical inputs in evaluation mode. “Mismatch px” counts pixels whose thresholded label changes. Δ IoU is likewise 0.0000 for every checkpoint.

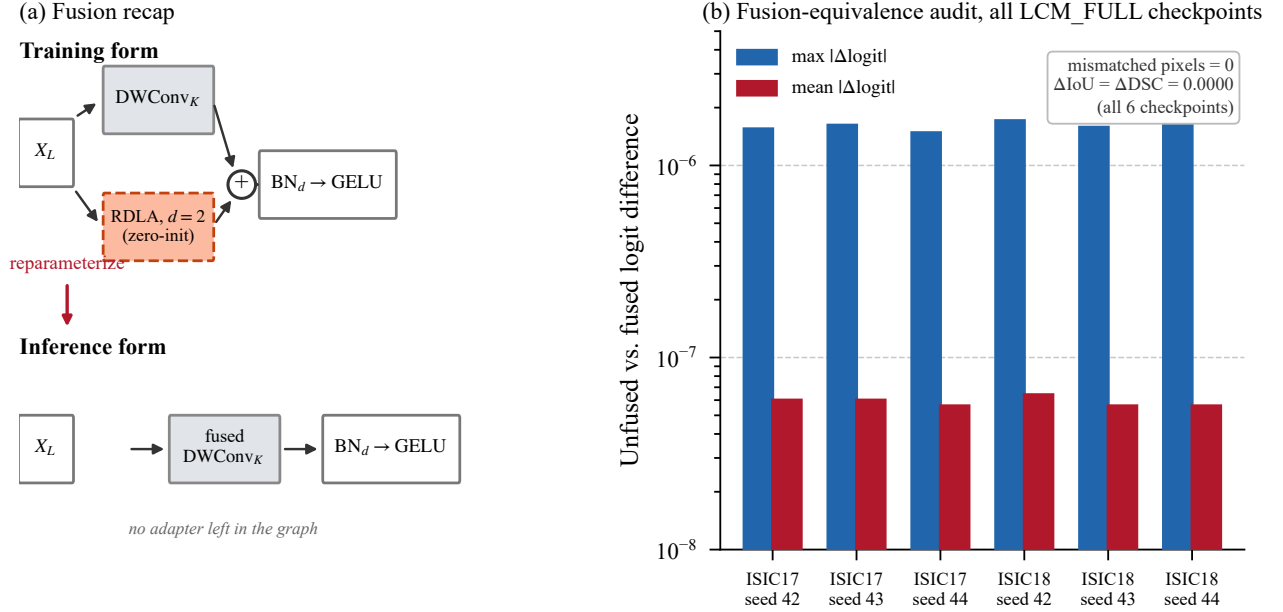


Fig. 13: (a) Compact recap of the training/inference fusion of Fig. 3: the host depthwise kernel and the zero-initialized RDLA branch are summed before BN_d during training, and the sum is folded into one kernel for inference, after which no adapter remains in the deployed graph. (b) The equivalence audit of Table XII plotted for all six LCM-UNet checkpoints: the maximum and mean absolute difference between unfused and fused logits is of order 10^{-6} and 10^{-8} respectively, consistent with single-precision floating-point rounding rather than an approximation, and every checkpoint produces zero mismatched thresholded pixels and zero change in IoU and DSC to four decimal places.

Two initialization properties were verified at model construction for every run. All adapter weight tensors contain exact zeros, so the training-form network reproduces the reference output bit-for-bit at step zero; and every adapter weight receives a non-zero gradient after the first backward pass, confirming that zero initialization does not freeze the branch.

J. Efficiency

Table XIII lists measured parameter counts and FLOPs under one profiling convention. The reference model has 34,176 parameters. Adding adapters raises the training-form count to 34,608, exactly the 432 weights predicted by Eq. (11); adding five skip scalars raises it to 34,612. After reparameterization the inference model contains 34,180 parameters, four more than the reference, and 0.05974 GFLOPs, identical to the reference to five decimal places. The adapters therefore cost 0.00038 GFLOPs during training and nothing at deployment.

Fig. 14 places these measurements alongside values reported in the literature for other lightweight dermoscopic segmentation models. Because those points come from their original publications rather than from our runs, they are drawn with open markers and must be read as reported values; the two filled markers are the only points measured under our protocol and are the only pair whose difference is controlled.

K. Comparison with Reported Results

Table XIV places LCM-UNet next to published numbers for comparable lightweight models. A protocol caveat applies to this comparison: those rows are transcribed from their source publications and were produced under each author’s own data

TABLE XIII: Measured Parameters and FLOPs at 256×256

Variant	Form	Parameters	GFLOPs
B0_REF	reference	34,176	0.05974
B1_RDLA	training	34,608	0.06012
B1_RDLA	fused inference	34,176	0.05974
B2_STAGE_SKIP	–	34,180	0.05974
LCM_FULL	training	34,612	0.06012
LCM_FULL	fused inference	34,180	0.05974

Counted from the instantiated models. FLOPs measured with `ptflops` 0.7.5 using one custom selective-scan counting hook applied identically to every locally profiled model. The fused model exceeds the reference by exactly the four stored skip scalars.

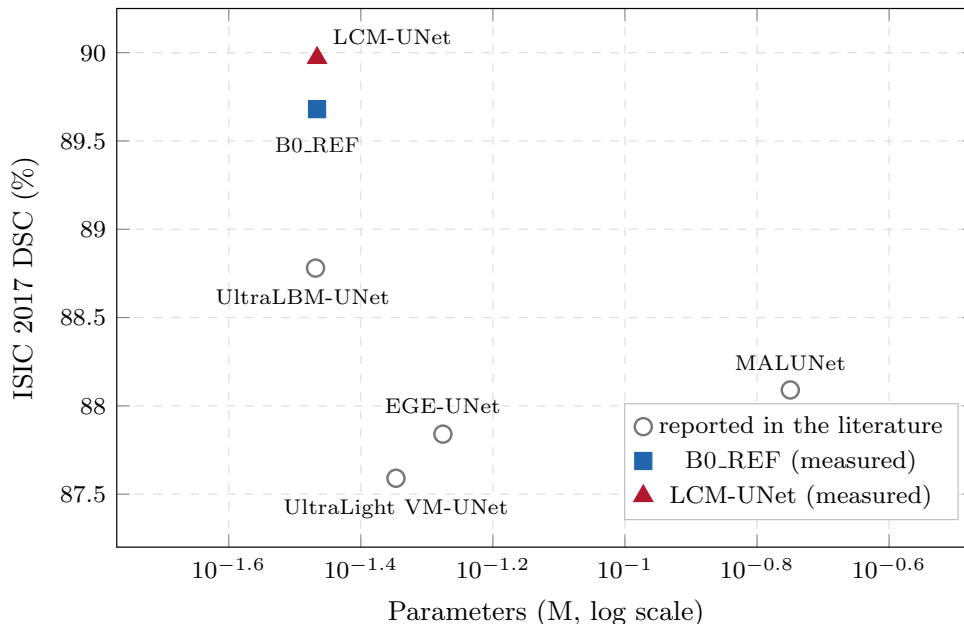


Fig. 14: DSC against parameter count on ISIC 2017 (log scale). Open markers are values reported in the original publications, shown for orientation only; filled markers are the two models measured here under identical splits, schedule, hardware and seeds, and are the only pair whose difference is controlled.

organization, training schedule and seed count, so the comparison is useful for orientation but is not controlled. The only controlled comparisons in this article are between B0_REF and LCM_FULL in Table VI and, for external data, in Table IX. It is also worth noting that our locally retrained B0_REF reaches 89.68% DSC on ISIC 2017, 0.90 points above the 88.78% published for the same architecture and data organization [19]. This gap exceeds the reference model’s own reported seed-wise standard deviation of ± 0.73 IoU, itself expressed on a different metric and not directly comparable in scale. We attribute the difference to the ordinary sources of cross-implementation variance at this parameter scale (framework and library versions, exact augmentation order, and other low-level details that a paper’s text cannot fully specify) rather than to any known deviation from the published recipe on our side, which we followed as described in Section IV-C. The gap is a further reason to treat Table XIV as context rather than as a controlled ranking, and it does not affect the B0_REF vs. LCM_FULL comparison, which uses one implementation, one set of splits and one schedule throughout.

Two contemporaneous designs illustrate why we frame this as context rather than as a race against every published number. MUCM-Net [43] and LiteMamba-Bound [44] report DSC on ISIC 2017 and ISIC 2018 that is competitive with the numbers in Table XIV, but the former is a UCM-Net variant with its own data split and training recipe and the latter operates at roughly 0.96 M parameters – almost thirty times the budget of every model actually compared in this article – so neither number is transcribed into Table XIV, and neither would be a controlled comparison against LCM-UNet if it were. This is the same caution that applies to MALUNet, EGE-UNet and UltraLight VM-UNet already listed there: at the parameter and FLOP scale this article targets, differences of a few tenths of a DSC point between independently implemented systems are within the range that framework version, augmentation order and random initialization alone can produce, so a same-implementation controlled comparison (Table VI) remains the only comparison in this article we treat as decisive.

TABLE XIV: Context Against Reported Lightweight Models on ISIC 2017

Model	Source	Params (M)	GFLOPs	DSC (%)
MALUNet [9]	reported	0.178	0.083	88.09
EGE-UNet [10]	reported	0.053	0.072	87.84
UltraLight VM-UNet [18]	reported	0.045	0.069	87.59
UltraLBM-UNet [19]	reported	0.034	0.060	88.78
B0_REF (ours)	measured	0.0342	0.0597	89.68
LCM-UNet (ours)	measured	0.0342	0.0597	89.97

Rows marked *reported* are transcribed from the cited publications and were produced under each author’s own data organization and training schedule; they are shown for orientation and are not a controlled comparison. Only the two *measured* rows share splits, schedule, hardware and seeds.

L. Efficiency-Accuracy Trade-off Analysis

Fig. 15 redraws the comparison of Table XIV as a GFLOPs-vs-IoU trade-off plot, extending it with two heavier literature baselines, U-Net and UNeXt, taken from the same source table in UltraLBM-UNet’s own publication [19] that supplies the other reported rows. Plotting cost on a log scale makes the scale gap explicit: LCM-UNet and B0_REF sit roughly two orders of magnitude below U-Net in GFLOPs while reaching higher IoU, and the ultra-light cluster itself (MALUNet, EGE-UNet, UltraLight VM-UNet, UltraLBM-UNet, and the two models measured here) spans less than a factor of two in cost at the right-hand edge of that gap. Within that cluster, LCM-UNet and B0_REF occupy the same GFLOPs as UltraLBM-UNet while reaching the highest IoU of the group, consistent with the efficiency accounting of Section V-J: the adapter changes the training-time parameter count but not the deployed GFLOPs. As with Table XIV, only the B0_REF-vs-LCM-UNet pair is a controlled comparison; the remaining points are reported values included for orientation.

M. Qualitative Results

Fig. 16 shows representative cases spanning a median-difficulty lesion, an irregular low-contrast boundary, a hair-occluded image, a small lesion, a failure case and an external PH² image, complementing the four-case, boundary-focused view already given in Fig. 8 with a wider gallery and, importantly, a case where the method does not help. The improvements are localized: on the irregular-boundary and hair-occlusion cases LCM-UNet follows concavities that the reference smooths over, and it produces fewer isolated false-positive fragments in hair-dense regions. On the small-lesion case both models under-segment, and on the failure case both miss a large portion of a lesion whose border is almost invisible under strong illumination gradient; the proposed components do not repair that class of error. This is consistent with the size of the measured effect: the adapter refines contours that are already approximately correct, and it does not change the outcome when the region-level decision itself fails.

N. Confusion-Matrix Record

Table XV lists the raw pixel counts from which every overlap metric in this article is derived, including the B0_REF PH² rows that make Table IX a controlled comparison, so that all reported values can be recomputed independently through Eqs. (15)–(17).

VI. DISCUSSION

A. What the Results Establish

Four findings are supported by the evidence in Section V. First, a zero-initialized dilated depthwise adapter placed in the local branch of a global-local state space block yields a consistent, if small, accuracy improvement in a network of only 34k parameters. The direction of the effect is preserved across two datasets, three seeds each, four ablation configurations and twelve external checkpoints (both models on PH²), and the paired image-level analysis shows that it is distributed across the evaluation set rather than concentrated in a few images. Second, the improvement is larger, in relative terms, on boundary-sensitive measures than on region overlap, which matches the mechanism the component was designed to address; isolating B1_RDLA in the boundary evaluation shows that the adapter alone is responsible for most of this gain. Third, the improvement measured in-distribution on ISIC 2017 and ISIC 2018 reappears at closely matched magnitude on PH², an external dataset seen by neither model during training, with the reference model now evaluated under the identical protocol so the transfer comparison is controlled rather than a single-model measurement. Fourth, the reparameterization is exact: no checkpoint produced a single changed pixel after fusion, so whatever accuracy is gained is obtained at a deployment cost of four scalars.

The magnitude is the part that must not be overstated. Under a seed-paired test the improvement is significant on both datasets ($p = 0.001$ on ISIC 2017, $p = 0.034$ on ISIC 2018); under the more conservative unpaired test at three seeds it is significant on ISIC 2017 ($p = 0.030$) but not on ISIC 2018 ($p = 0.120$). The image-level analysis, with hundreds of paired observations per dataset, is significant at $p < 10^{-4}$ on both datasets, with rank-biserial correlations of 0.19–0.20. The defensible summary is a small but statistically well-supported shift, and the practical value of the finding lies as much in its cost as in its size: it is obtained without adding an inference-time operator.

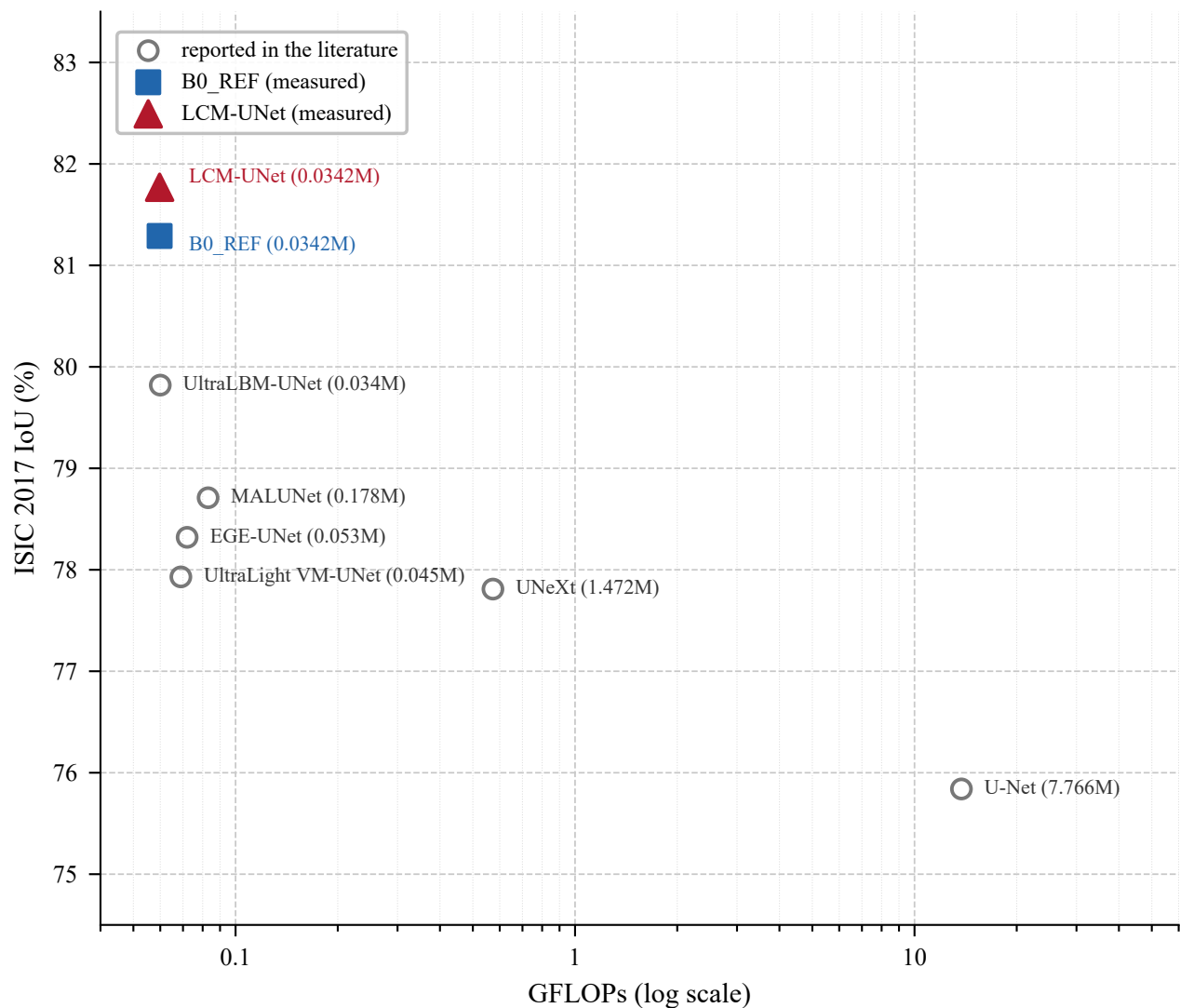


Fig. 15: ISIC 2017 IoU against GFLOPs (log scale), extending Fig. 14 with two additional literature points (U-Net, UNeXt) and an IoU rather than a DSC axis. Open markers are values reported in UltraLBM-UNet’s own comparison table [19], shown for orientation only; filled markers are the two models measured in this article under identical splits, schedule, hardware and seeds. B0_REF and LCM-UNet sit at the same GFLOPs as UltraLBM-UNet while reaching higher IoU, and both are roughly two orders of magnitude cheaper than U-Net. TransUNet and Swin-UNet, named alongside these baselines in Table I, are not plotted here because neither their original publications nor UltraLBM-UNet’s comparison table report GFLOPs/IoU for them on ISIC 2017 or ISIC 2018; their canonical benchmarks are Synapse multi-organ CT and ACDC.

B. Why the Adapter Helps

Because fusion is exact, the training-form and inference-form networks realize the same function class. The adapter therefore cannot enlarge the hypothesis space of the deployed model; what it changes is the optimization path used to reach a point inside it. Two effects plausibly contribute. The adapter supplies a second, sparser sampling lattice whose gradients reach a subset of the host kernel taps directly, which alters the conditioning of the depthwise update in a branch that has only 25 or 49 weights per channel. Zero initialization means this alternative path is switched on gradually rather than perturbing an otherwise well-behaved starting point.

One finding complicates a simple version of this story: LCM-UNet does not have a smaller seed-wise standard deviation than the reference on every metric. The two models are close on ISIC 2017, and on ISIC 2018 LCM-UNet is visibly more variable across seeds, driven by one high-scoring run (Table VI). Whatever the adapter changes about the optimization path, it does not act as a general variance-reduction mechanism, so repeatability is not part of the case made for the method here; the case rests on the mean shift and its consistency across datasets, ablations and the paired image-level test, not on any claim about seed-to-seed stability.

The mechanism described above is an interpretation of the measured behaviour, not something the present experiments prove

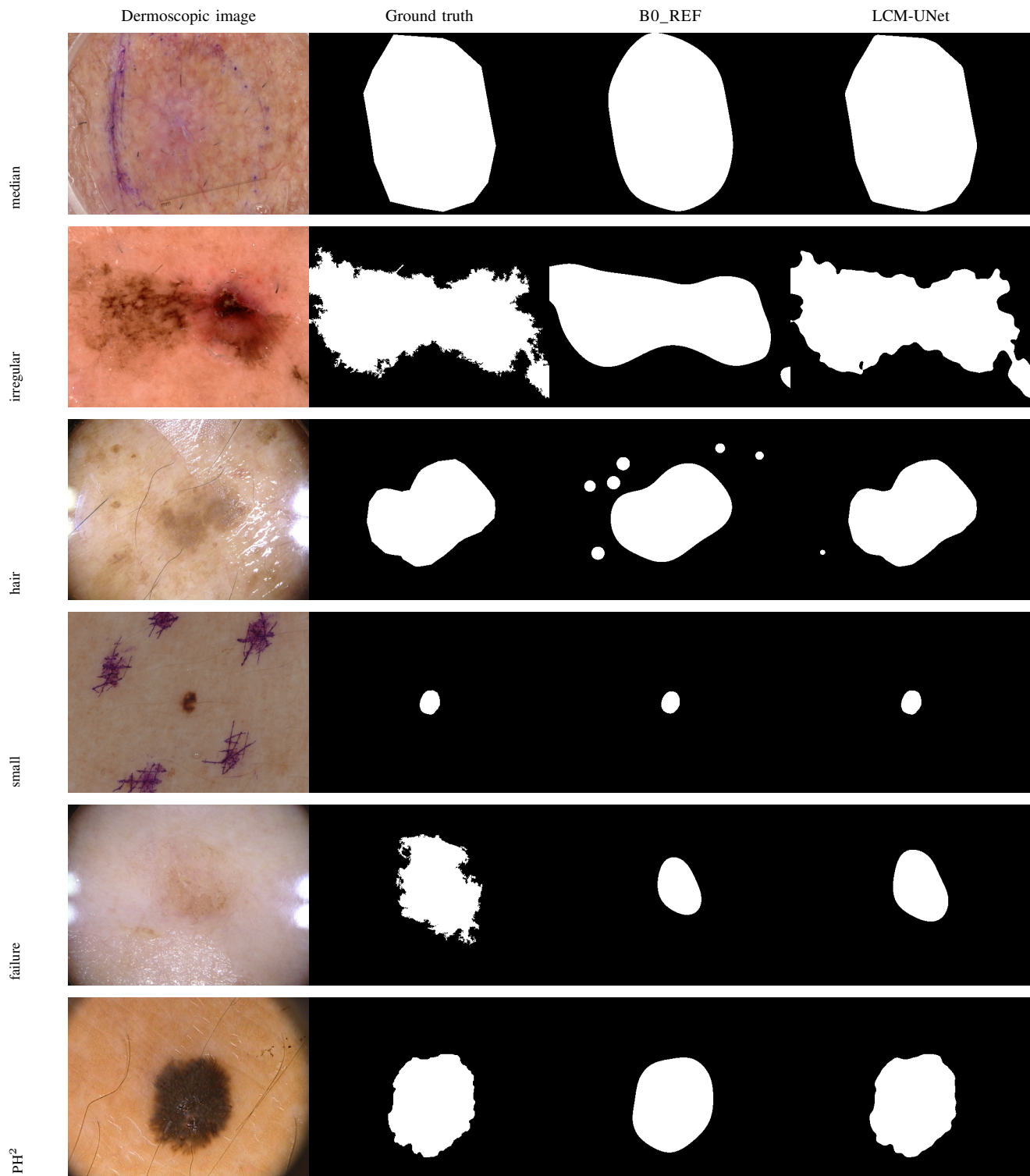


Fig. 16: Qualitative comparison on six representative cases, from a median-difficulty lesion to a failure case and an external PH² image (seed-43 checkpoints, threshold 0.5). LCM-UNet follows concavities the reference smooths over and leaves fewer isolated fragments in hair-dense regions; neither model recovers the failure case, where the border is largely erased by an illumination gradient.

outright. Section VI-F describes the decisive test that would settle it, whose outcome we do not anticipate here.

TABLE XV: Pixel-Pooled Confusion Matrices Underlying All Reported Overlap Metrics

Run	Dataset	Seed	TN	FP	FN	TP	IoU (%)	DSC (%)
B0_REF	ISIC 2017	42	30,044,896	1,382,297	928,595	9,914,932	81.10	89.56
B0_REF	ISIC 2017	43	30,052,363	1,374,830	896,094	9,947,433	81.41	89.75
B0_REF	ISIC 2017	44	30,050,467	1,376,726	901,374	9,942,153	81.36	89.72
LCM-UNet	ISIC 2017	42	30,048,676	1,378,517	875,570	9,967,957	81.56	89.84
LCM-UNet	ISIC 2017	43	30,056,305	1,370,888	841,956	10,001,571	81.88	90.04
LCM-UNet	ISIC 2017	44	30,055,076	1,372,117	842,631	10,000,896	81.87	90.03
B0_REF	ISIC 2018	42	38,697,731	1,490,444	1,188,015	11,576,898	81.21	89.63
B0_REF	ISIC 2018	43	38,707,530	1,480,645	1,148,167	11,616,746	81.55	89.84
B0_REF	ISIC 2018	44	38,702,995	1,485,180	1,167,787	11,597,126	81.38	89.74
LCM-UNet	ISIC 2018	42	38,684,154	1,504,021	1,135,389	11,629,524	81.50	89.81
LCM-UNet	ISIC 2018	43	38,698,300	1,489,875	1,067,929	11,696,984	82.06	90.14
LCM-UNet	ISIC 2018	44	38,689,758	1,498,417	1,113,006	11,651,907	81.69	89.92
B0_REF	PH ² (from ISIC 2017)	42	9,539,931	279,628	238,288	3,049,353	85.48	92.17
B0_REF	PH ² (from ISIC 2017)	43	9,539,931	279,628	222,902	3,064,739	85.91	92.42
B0_REF	PH ² (from ISIC 2017)	44	9,539,931	279,628	229,214	3,058,427	85.74	92.32
B0_REF	PH ² (from ISIC 2018)	42	9,556,502	263,057	238,288	3,049,353	85.88	92.40
B0_REF	PH ² (from ISIC 2018)	43	9,556,502	263,057	222,902	3,064,739	86.31	92.65
B0_REF	PH ² (from ISIC 2018)	44	9,556,502	263,057	229,214	3,058,427	86.14	92.55
LCM-UNet	PH ² (from ISIC 2017)	42	9,558,974	260,585	238,288	3,049,353	85.94	92.44
LCM-UNet	PH ² (from ISIC 2017)	43	9,558,974	260,585	222,902	3,064,739	86.37	92.69
LCM-UNet	PH ² (from ISIC 2017)	44	9,558,974	260,585	229,214	3,058,427	86.20	92.59
LCM-UNet	PH ² (from ISIC 2018)	42	9,575,369	244,190	238,288	3,049,353	86.34	92.67
LCM-UNet	PH ² (from ISIC 2018)	43	9,575,369	244,190	222,902	3,064,739	86.77	92.92
LCM-UNet	PH ² (from ISIC 2018)	44	9,575,369	244,190	229,214	3,058,427	86.60	92.82

Counts are pooled over every evaluation pixel at threshold 0.5. IoU and DSC in the last two columns are obtained from these counts through Eq. (15); all aggregate values elsewhere in the article can be reconstructed from this table. PH² rows report both B0_REF and LCM-UNet checkpoints, so the external validation in Table IX is, for the first time in this study, a controlled comparison rather than a single-model transfer measurement.

C. Comparison with Prior Reparameterization Work

The kernel identity used here is prior art [26], as set out in Section III-E, and we claim no novelty for it. What this article contributes is a measurement of that mechanism in a regime where it had not been evaluated. Prior reparameterization studies operate on classification and detection backbones of 10–400 M parameters, where a host kernel holds hundreds to thousands of weights and three to six parallel branches across several kernel sizes and dilation rates are typical. Our host kernels hold 25 or 49 weights per channel, sit inside a channel group of eight to sixteen channels, and share a block output with a selective-scan branch. In that regime we find that a single dilated branch without its own normalization is sufficient, and that the resulting effect is real but modest, on the order of a quarter to a third of a DSC point in-distribution and of similar size on fully external data. That is a useful calibration for practitioners: the technique transfers to ultra-light state space segmenters, but its magnitude here is far smaller than the improvements reported on large convolutional backbones, and it should be budgeted accordingly rather than assumed.

D. Computational Deployment Considerations

The efficiency numbers in Section V-J are worth translating into what they would mean for the point-of-care hardware this model class targets, while being explicit about what is and is not established by doing so. At 34,180 parameters, storing the reparameterized model requires under 134 KB in 32-bit floating point and well under 40 KB if quantized to 8-bit integers, small enough to fit in the on-chip SRAM of many low-power microcontrollers rather than requiring external memory access, which is often the dominant energy cost on such devices. At 0.05974 GFLOPs per 256×256 inference, the arithmetic cost is roughly two to three orders of magnitude below the transformer-based segmenters cited in Section II and comparable to the other ultra-light Mamba-based networks in Table XIV, so a device capable of running UltraLBM-UNet at a given frame rate should, in principle, run LCM-UNet at an indistinguishable rate: the adapter is removed before inference and the four skip scalars are ordinary multiply-adds folded into existing tensors. We stress *in principle* because FLOPs are a poor proxy for measured latency on selective-scan kernels specifically, whose throughput depends on how well a given runtime fuses the scan into hardware-efficient kernels, a detail that varies across mobile NPUs, microcontroller DSPs and general-purpose embedded CPUs. We did not benchmark on any such target, so this subsection should be read as a plausibility argument from the algebra of the reparameterization, not as a deployment result; Section VI-F lists on-device latency and memory measurement as the first item that would convert this argument into evidence.

E. Broader Impact and Clinical Translation

The intended use of this work is as a preprocessing or triage aid for dermoscopic screening on constrained hardware, not as an autonomous diagnostic decision-maker, and that distinction matters for how the results should be read outside a benchmark setting. A boundary mask, on its own, does not diagnose anything; it is one input to downstream measurements (asymmetry, border irregularity, diameter) that a clinician or a classification model would still need to interpret, so the appropriate deployment context is one with a human, or a separately validated classifier, in the loop. Two considerations follow from this.

First, none of ISIC 2017, ISIC 2018 or PH² documents the Fitzpatrick skin-type distribution of its images, and the broader dermoscopic-imaging literature has repeatedly found that public benchmark collections of this kind are skewed toward lighter skin tones, with corresponding drops in diagnostic performance on darker skin that a benchmark restricted to the same skewed distribution cannot reveal [62]. Because LCM-UNet is trained and evaluated exclusively on these three sources, the accuracy figures in Section V characterize performance on the population those datasets represent and should not be read as evidence of uniform performance across skin tones; we are not aware of a skin-type annotation for any of the three datasets that would let us measure this gap directly; a fully mitigating fix (rebalanced training data, or fairness-aware training) is outside the architectural scope of this article, and we flag it as an unaddressed dimension of external validity alongside the flat train/evaluation split of Section VI-F rather than fold it into that paragraph, since the two limitations have different remedies.

Second, the efficiency argument of Section VI-D is what makes low-cost, local, offline screening feasible in settings without reliable connectivity to a cloud-hosted model; that accessibility case is precisely why an unaudited fairness gap would be consequential rather than a purely academic caveat, since the populations most likely to depend on an offline point-of-care tool are not, in general, the ones best represented in existing dermoscopic archives. We report this as a scope boundary of the evaluation, to be weighed by anyone considering deployment, rather than as a property the architecture itself could be expected to fix.

F. Limitations

Four limitations bound the conclusions, and we state each together with the experiment that would resolve it.

Capacity-matched control. The adapter adds 432 trainable weights during optimization. Attributing the effect specifically to the sparse dilated sampling lattice, rather than to the extra training-time capacity or to the presence of a parallel zero-initialized branch of any geometry, requires a control branch with identical parameter count and dilation one, merged into the centre 3×3 of the host kernel. That experiment is not part of this study; it is, in our view, the single most informative addition this line of work could receive next.

Statistical power. Three seeds support a paired run-level test on both datasets and an unpaired run-level test on ISIC 2017, but not an unpaired test on ISIC 2018. Five seeds per configuration would make the run-level comparison reportable in its own right on every dataset instead of relying, for ISIC 2018, on the paired test and on the image-level analysis.

Model selection protocol. As stated in Section IV-C, the held-out partition serves as both validation and evaluation set, following the convention of the reference line of work. The comparison between variants is unaffected because the procedure is identical for all of them, but absolute values are optimistic relative to a three-way split and should not be compared across protocols.

Efficiency reporting and domain scope. Parameters and FLOPs are measured, but latency and peak memory on an embedded target are not. FLOPs are an imperfect proxy for on-device speed, particularly for selective-scan kernels, so the zero-cost claim is established algebraically and in FLOPs but not yet on hardware (Section VI-D). All experiments are binary two-dimensional dermoscopic segmentation at 256×256 ; multi-class, volumetric and higher-resolution behaviour is untested.

Ground-truth reliability. Every metric in this article is computed against a single consensus mask per image, but dermoscopic boundary annotation is known to carry substantial inter-annotator disagreement, particularly for lesions with diffuse or ill-defined borders, and that disagreement correlates with exactly the malignancy-relevant cases a clinical system would most need to get right [63]. A boundary-sensitive gain of a few tenths of a Boundary Dice point (Section V-G) is measured against a label that is itself uncertain by a margin of comparable or larger size on the hardest images, which this study does not quantify because none of ISIC 2017, ISIC 2018 or PH² ships multiple independent annotations per image. Reporting against a multi-annotator boundary reference, where available, would place the measured effect size in the context of label noise rather than treating the single provided mask as ground truth in the literal sense. The failure case in Fig. 16 is a concrete instance of exactly this difficulty: a lesion whose border is almost erased by an illumination gradient, where both models miss a large portion of the region and neither the adapter nor the skip refinement changes the outcome. We include it deliberately, alongside the four working examples of Fig. 8, because the case for LCM-UNet in this article rests on refining boundaries that are already approximately correct, not on repairing region-level failures of this kind, and a qualitative figure that showed only favourable cases would overstate that case.

VII. CONCLUSION

This article presented LCM-UNet, a local-compensated Mamba U-Net for lightweight skin lesion segmentation. The network strengthens the narrow local pathway of a global-local state space block with a Residual Dilated Local Adapter: a zero-initialized, dilation-two depthwise 3×3 convolution summed into the pre-activation depthwise response of four GLMBP

blocks. Because the sum precedes the first normalization and both branches are bias free, the adapter is merged exactly into the host kernel after training and disappears from the deployed graph. A minor refinement replaces the single shared outer skip coefficient with five stage-specific scalars.

Across three seeds on ISIC 2017 and ISIC 2018, LCM-UNet improved over a locally retrained reference backbone by 0.48 and 0.37 IoU percentage points and 0.29 and 0.22 DSC percentage points, with lower HD95 and ASSD on both datasets. On PH², an external dataset seen by neither model during training, evaluating the reference backbone under the identical protocol showed that the gain transfers essentially unchanged: +0.46 IoU and +0.27 DSC points across six checkpoints per model, closely matching the in-distribution result. The improvement is significant under the seed-paired test on both datasets, under the unpaired three-seed test on ISIC 2017, and under the image-level test at $p < 10^{-4}$ on both datasets; the image-level effect sizes remain small by conventional benchmarks even though the tests themselves are decisive. We report the result as a consistent and statistically robust shift rather than a substantial advance in accuracy. The fusion audit confirmed that unfused and fused models produce identical thresholded masks to numerical tolerance, and the reparameterized model uses 34,180 parameters and 0.05974 GFLOPs, four parameters more than the reference at the same computational cost.

The wider contribution is a calibration rather than a claim of superiority. The dilated reparameterization equivalence was established on backbones three to four orders of magnitude larger; this article measures what it is worth at 0.034M parameters inside a selective-scan block, and finds a benefit that is real, consistent across datasets and external validation, and modest. Three directions follow directly. A dilation-one, capacity-matched control would separate sampling geometry from training-time capacity and is the decisive next experiment. Five seeds per configuration would make the run-level comparison reportable without relying on pairing. Finally, on-device latency and memory measurements would establish the zero-cost claim on hardware rather than in FLOPs alone.

APPENDIX A

EXTENDED DERIVATION OF THE DILATED KERNEL REPARAMETERIZATION IDENTITY

Section III-D states the fusion identity in the compressed form needed for the main narrative. This appendix gives the full derivation, including the case $K = 7$ that the main text only asserts, and states explicitly which properties of the network the identity depends on.

A. Dilated convolution as a sparse non-dilated convolution

Let X be a single feature-map channel (the argument is per-channel because every convolution in this article is depthwise) and let $\mathcal{A}$ be a 3×3 convolution with dilation d , no bias, and padding chosen so that output and input share spatial size. Writing weights as $W_{\mathcal{A}}[i, j]$ for $i, j \in \{-1, 0, 1\}$,

$$[\mathcal{A}(X)][p] = \sum_{i=-1}^1 \sum_{j=-1}^1 W_{\mathcal{A}}[i, j] X[p + d \cdot (i, j)], \quad (19)$$

where $p = (p_x, p_y)$ indexes spatial position and $p + d \cdot (i, j)$ is shorthand for $(p_x + di, p_y + dj)$. Now define a *non-dilated* convolution of size $k' = 2d + 1$ whose weights are zero except on the sub-lattice spaced by d :

$$W'[a, b] = \begin{cases} W_{\mathcal{A}}[a/d, b/d] & \text{if } a, b \in \{-d, 0, d\}, \\ 0 & \text{otherwise,} \end{cases} \quad a, b \in \{-d, \dots, d\}. \quad (20)$$

Because W' is zero everywhere except at the three-by-three sub-grid $\{-d, 0, d\}^2$,

$$[W' * X][p] = \sum_{a=-d}^d \sum_{b=-d}^d W'[a, b] X[p + (a, b)] = \sum_{i, j \in \{-1, 0, 1\}} W_{\mathcal{A}}[i, j] X[p + d(i, j)], \quad (21)$$

which is identical to (19) term by term, for every position p and every input X . The two operators are therefore the same function, not merely close in value; no approximation, truncation or interpolation is involved. Setting $d = 2$ gives $k' = 5$: the three sampled rows and columns fall at $\{-2, 0, 2\}$, which are indices $\{0, 2, 4\}$ once the kernel is written with a 0-based index running $0, \dots, 4$ – Eq. (7) of the main text with $K = 5$.

B. Embedding into a larger host kernel

The E6 and D1 blocks host a 7×7 depthwise kernel rather than a 5×5 one. The 5×5 sparse kernel from (20) must be embedded inside the larger 7×7 support, centred so that its own origin coincides with the host kernel's origin. A 7×7 kernel indexed $0, \dots, 6$ has its centre at index 3; embedding an offset $a \in \{-2, 0, 2\}$ at that centre places it at index $3 + a$, giving indices $\{1, 3, 5\}$ – the $K = 7$ case of Eq. (7). Formally, for $K \in \{5, 7\}$ define the index map $\pi_K(t) = \frac{K-1}{2} + dt$ for $t \in \{-1, 0, 1\}$, which specializes to $\pi_5(t) = 2t$ and $\pi_7(t) = 2t + 1$ once shifted to a 0-based index, matching the main text.

C. Why direct kernel addition is exact

Let W_K be the host kernel (size $K \times K$, non-dilated, depthwise, no bias) and let $\mathcal{E}_K(W_A)$ be the embedding of the adapter kernel constructed above. Convolution is linear in its kernel for a fixed input, stride and padding: for any two kernels W_K and W' of the same size applied with the same stride and padding, $(W_K * X) + (W' * X) = (W_K + W') * X$. The host branch and the embedded adapter branch share input X_L , stride 1, dilation 1 (after embedding) and the padding needed to preserve spatial size, so

$$Z = [W_K * X_L] + [\mathcal{E}_K(W_A) * X_L] = [W_K + \mathcal{E}_K(W_A)] * X_L = W_{\text{fused}} * X_L, \quad (22)$$

which is Eq. (8). Because both branches are bias-free and this identity holds pointwise in X_L before any nonlinearity is applied, Z is bit-identical (up to floating-point summation order) whether it is produced by the two-branch training-form network or by the single fused kernel. Every downstream operation – BN_d , the GELU nonlinearity, the pointwise convolution, and the rest of the network – is then a function of the same Z in both forms, which is why no other parameter needs to change and why the equivalence is exact rather than approximate: the argument never leaves the domain of linear operators until after the point where the two forms are shown to agree.

D. Consequence at initialization

Setting $W_A = \mathbf{0}$ in (20) gives $\mathcal{E}_K(\mathbf{0}) = \mathbf{0}$, so $W_{\text{fused}} = W_K$ at initialization: the zero-initialized training-form network and the reference network compute the same Z , and therefore the same output, before a single optimization step has been taken (Eq. (6)). The derivation above shows this is not a special property that has to be separately verified – it is the $W_A = \mathbf{0}$ case of the same linear identity that justifies fusion after training, which is why one audit (Algorithm 2) suffices to check both properties on a trained checkpoint: a zero maximum logit difference after fusion confirms the general identity, and the zero-initialization check in Section V-I confirms the boundary case independently, from the stored initial weights rather than from the identity itself.

APPENDIX B

WHY ZERO INITIALIZATION, RATHER THAN AN ALTERNATIVE

Section III-C motivates zero initialization by appeal to residual-scaling and zero-initialized adaptation schemes [46]–[49]; this appendix makes the argument for *this* construction self-contained rather than only citing the precedent.

Three initializations are worth contrasting. *Random initialization* (the default for a freshly instantiated convolution) would make $W_{\text{fused}} \neq W_K$ at step 0, so the training-form network would not compute the same function as the reference architecture before any optimization has occurred; whatever gain or loss is subsequently measured would be confounded with a different starting point, which is precisely the ambiguity Section III-C rules out by construction. *Small-magnitude random initialization* (drawing W_A from a narrow distribution rather than exactly zero) softens but does not remove the same problem: Z at step 0 would differ from the reference by an amount proportional to the initialization scale, so the function-identity property of Eq. (6) becomes approximate rather than exact, and "approximately the same starting point" is a weaker and harder-to-audit claim than "the same starting point," which the zero-initialization case lets us verify by inspecting stored weights rather than by comparing outputs on a held-out batch. *Zero initialization*, by contrast, makes the training-form and reference networks compute bit-identical outputs at step 0 as a direct consequence of the linearity argument in Appendix A, independent of the data the network will subsequently see; the only thing that needs to be verified empirically is that gradients still reach W_A , since a badly chosen zero-initialized branch could in principle sit at a saddle point of its own local loss landscape. Section V-I reports that every adapter weight in every trained checkpoint receives a non-zero gradient after the first backward pass, which rules out that failure mode in practice for this architecture. This is also why Algorithm 2 audits both properties together on every trained model rather than treating zero-initialization as a design choice that, once made, requires no further checking: the identity guarantees the starting point, but only training verifies that the branch was actually used.

APPENDIX C

WORKED NUMERICAL EXAMPLE OF THE FUSION IDENTITY

The indices in Eq. (7) are easiest to check against a concrete example rather than the general formula alone. Take an illustrative (not a trained) 3×3 adapter kernel for one channel,

$$W_A = \begin{pmatrix} 0.10 & -0.05 & 0.02 \\ 0.03 & 0.20 & -0.04 \\ -0.01 & 0.06 & 0.08 \end{pmatrix}, \quad (23)$$

and an illustrative 5×5 host kernel W_5 with all entries equal to 0.01 for readability. Embedding $W_{\mathcal{A}}$ at rows/columns $\{0, 2, 4\}$ per $\pi_5(a) = 2a$, $a \in \{0, 1, 2\}$ (Eq. (7)) gives

$$\mathcal{E}_5(W_{\mathcal{A}}) = \begin{pmatrix} 0.10 & 0 & -0.05 & 0 & 0.02 \\ 0 & 0 & 0 & 0 & 0 \\ 0.03 & 0 & 0.20 & 0 & -0.04 \\ 0 & 0 & 0 & 0 & 0 \\ -0.01 & 0 & 0.06 & 0 & 0.08 \end{pmatrix}, \quad (24)$$

and the fused kernel $W_{\text{fused}} = W_5 + \mathcal{E}_5(W_{\mathcal{A}})$ replaces every entry at an even row and even column with 0.01 plus the corresponding adapter weight, and leaves every entry with at least one odd index (the sixteen positions the dilated 3×3 footprint never touches) equal to the untouched host value 0.01. A dilated convolution with 9 multiply-adds per channel and a dense 5×5 convolution with 25 multiply-adds per channel therefore become, after fusion, a single dense 5×5 convolution with 25 multiply-adds and no separate adapter pass – the 16 positions where $\mathcal{E}_5$ is zero cost nothing extra at inference because they were already part of the host kernel’s 25 weights before the adapter existed.

APPENDIX D STATISTICAL METHODOLOGY IN DETAIL

Section IV-D and Section V-H state the statistical procedures at the level needed to interpret their outcomes; this appendix gives the exact formulas, so that every p -value and interval in Tables VII and XI can be recomputed from the confusion-matrix archive (Table XV) or the per-image Dice values without ambiguity about convention.

A. Run-level Student t -tests

Let $a_1, \dots, a_n$ and $b_1, \dots, b_n$ ($n = 3$) be the seed-wise values of a metric for two compared variants. The unpaired, two-sided test uses Welch’s statistic with unequal variances assumed,

$$t_{\text{unpaired}} = \frac{\bar{b} - \bar{a}}{\sqrt{s_a^2/n + s_b^2/n}}, \quad (25)$$

with $\bar{a}, \bar{b}$ the sample means and s_a^2, s_b^2 the sample variances (Bessel’s correction, divisor $n - 1$), and degrees of freedom from the Welch–Satterthwaite equation. The seed-paired test instead forms the per-seed differences $\delta_i = b_i - a_i$ for matched seed i and computes

$$t_{\text{paired}} = \frac{\bar{\delta}}{s_{\delta}/\sqrt{n}}, \quad (26)$$

with $n - 1$ degrees of freedom. Both are evaluated two-sided at $\alpha = 0.05$; we report the exact p -value rather than only the accept/reject decision, and both variants are given side by side in every case (Table VII) rather than only the one that is significant.

B. Wilcoxon signed-rank test and rank-biserial correlation

For the paired image-level analysis, let $\delta_1, \dots, \delta_n$ be the per-image Dice differences (LCM-UNet minus B0_REF, each already averaged over the three seeds, so $n = 645$ for ISIC 2017 and $n = 808$ for ISIC 2018). Discarding exact zeros, rank the absolute differences $|\delta_i|$ and let W_+ and W_- be the sums of ranks corresponding to positive and negative δ_i respectively. The Wilcoxon signed-rank statistic is $W = \min(W_+, W_-)$, and its p -value is obtained from the normal approximation with continuity correction at this sample size. The effect size we report is the matched-pairs rank-biserial correlation,

$$r_{\text{rb}} = \frac{W_+ - W_-}{W_+ + W_-} \in [-1, 1], \quad (27)$$

which is 0 when positive and negative ranks are balanced and ± 1 when every difference has the same sign; it is reported instead of, say, Cohen’s d because it makes no distributional assumption about δ_i , consistent with using a rank test in the first place.

C. Bootstrap confidence interval

The 95% confidence interval for the mean paired difference $\bar{\delta}$ is a percentile-bootstrap interval: draw $B = 10,000$ resamples of size n from $\{\delta_1, \dots, \delta_n\}$ with replacement, compute the resample mean $\bar{\delta}^{(1)}, \dots, \bar{\delta}^{(B)}$, and report the 2.5th and 97.5th percentiles of that empirical distribution as the interval bounds. This makes no normality assumption on δ_i and is consistent, by construction, with the paired (not independent) sampling unit being the image rather than the seed $\times$ image combination, matching the averaging step described in Section IV-D.

TABLE XVI: Variant Configuration Summary

Variant	RDLA	Skip	Train params	Infer. params	Datasets
B0_REF	–	shared	34,176	34,176	ISIC17, ISIC18
B1_RDLA	✓	shared	34,608	34,176	ISIC17
B2_STAGE_SKIP	–	per-stage	34,180	34,180	ISIC17
LCM_FULL	✓	per-stage	34,612	34,180	ISIC17, ISIC18

“Skip” is the outer skip-scaling scheme (Sec. III-F); “Train/Infer. params” are total model parameters before/after reparameterization (Sec. III-D); “Datasets” lists which of ISIC 2017/2018 each variant was trained on (Sec. IV-B). PH² external validation (Table IX) uses only the B0_REF and LCM_FULL checkpoints trained on each ISIC release.

TABLE XVII: Reproducibility Checklist

Item	Status	Where
Model architecture fully specified	Yes	Sec. III
Reference implementation released	Yes	<code>codes/model/</code>
Training hyperparameters specified	Yes	Table V
Data splits reproducible (seeded, hashed)	Yes	Sec. IV-A
Number of seeds per configuration	Yes ($n=3$)	Sec. IV-C
Compute environment specified	Yes	Sec. IV
Evaluation metrics formally defined	Yes	Eq. (15)–(18)
Statistical tests and assumptions stated	Yes	Sec. IV-D
Variance/uncertainty reported	Yes (SD, 3 seeds; bootstrap CI, image-level)	Sec. V
Ablations isolate each component	Yes	Sec. V-E
Trained checkpoints released	Not yet	Data Availability
Hyperparameter search described	No – reference recipe adopted unmodified, not tuned	Sec. IV-C
Capacity-matched control for RDLA	No – identified as the decisive next experiment	Sec. VI-F
On-device latency/memory measured	No – FLOPs/parameters only	Sec. VI-D
Dataset licenses confirmed reusable	Yes – all three sources are public research benchmarks	Data Availability

APPENDIX E

VARIANT CONFIGURATION SUMMARY

Section IV-B introduces the four compared configurations by flag; Table XVI collects their parameter counts and dataset coverage in one place for reference, since these details are otherwise scattered across Sections III-G, IV-B and V-J.

APPENDIX F

REPRODUCIBILITY CHECKLIST

Table XVII summarizes what is and is not fixed by this article well enough for independent reproduction, following the spirit of the reproducibility checklists now common at NeurIPS and MICCAI. Entries marked *partial* or *no* are cross-referenced to the limitation or future-work item that explains the gap rather than left unexplained.

APPENDIX G

NOTATION AND ABBREVIATIONS

Table XVIII collects the symbols used across Sections III–V for reference, and Table XIX the abbreviations used throughout; each is also defined or expanded at first use in the main text.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY

All datasets used in this study are publicly available: ISIC 2017 [1], ISIC 2018 [2] and PH² [53]. No new data were generated. Code is available at <https://github.com/Shihabul-Shuvo/LCM-UNet>; trained model checkpoints will be released at the same repository upon publication.

TABLE XVIII: Notation

Symbol	Meaning
X	Input image, $3 \times H \times W$
C, H', W'	Channel count and spatial size at a given stage
N	Token count, $N = H'W'$
LN, Reshape	Layer normalization; flatten to token sequence
X_1, X_2, X_L, X_I	The four GLMBP channel groups (global, global, local, identity)
$\mathcal{M}$	The shared bidirectional Mamba module
Flip	Reverses token order for the backward scan
s_j	Internal residual scale of global group $j \in \{1, 2\}$
DWConv $_K$	Depthwise convolution, kernel size K
BN $_d$, BN $_p$	Depthwise-branch and pointwise-branch batch normalization
PW	1×1 pointwise convolution
σ	GELU activation
Z	Pre-activation depthwise response, $Z = \text{DWConv}_K(X_L)$
$\mathcal{A}, W_{\mathcal{A}}$	The RDLA adapter operator and its 3×3 weight tensor
d	Adapter dilation ($d = 2$ throughout)
K	Host kernel size, $K \in \{5, 7\}$
$\mathcal{E}_K(\cdot)$	Embedding of the 3×3 adapter kernel into a $K \times K$ sparse kernel
$\pi_K(\cdot)$	Index map used by $\mathcal{E}_K$ (Appendix A)
W_K, W_{fused}	Host kernel; host + embedded-adapter kernel after fusion
θ, θ'	Training-form and inference-form (fused) parameter sets
κ_j	Outer skip scalar for level $j \in \{1, \dots, 5\}$
$E_j, \hat{D}_j, D_j$	Encoder feature, upsampled decoder feature, and fused decoder feature at level j
$\mathcal{L}_{\text{BCE}}, \mathcal{L}_{\text{Dice}}$	Binary cross-entropy and Dice loss terms
P, Y	Predicted probability map; ground-truth mask
TP, TN, FP, FN	Pixel-pooled confusion-matrix counts at threshold 0.5
IoU, DSC, Acc, Sen, Spe	Intersection-over-union, Dice, accuracy, sensitivity, specificity
r_{rb}	Matched-pairs rank-biserial correlation (Sec. V-H)

TABLE XIX: Abbreviations

Abbrev.	Meaning
ASSD	Average symmetric surface distance
BCE	Binary cross-entropy
BN	Batch normalization
CI	Confidence interval
DSC	Dice similarity coefficient
FLOPs	Floating-point operations
GELU	Gaussian error linear unit
GLMBP	Global-local multi-branch perception (block)
HD95	95th-percentile Hausdorff distance
IoU	Intersection over union
LMBP	Local multi-branch perception (block)
LN	Layer normalization
NPU	Neural processing unit
PH ²	Pedro Hispano Hospital dermoscopy database
RDLA	Residual Dilated Local Adapter
S4 / S6	(Selective) structured state space sequence model
SD	Standard deviation
SSM	State space model

ETHICS STATEMENT

This study used only publicly available, fully de-identified image datasets. No new patient data were collected and no institutional review board approval was required.

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